Article

A Simple and Rigorous Field Interpretation of Dark Matter Phenomenology

Universal Quantum Foam Hypothesis

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Abstract

This work develops and tests a simple, reproducible, and rigorous field interpretation of dark matter phenomenology. The dark matter effect is treated as a possible nonlocal, scale-dependent, and saturation-limited field response to baryonic organization. The galactic analysis is based on 175 SPARC galaxies with 3391 radial measurement points. The central target quantity is the positive acceleration excess,

$$g_{\mathrm{field,obs}} = g_{\mathrm{obs}} - g_{\mathrm{bar}}, \qquad g_{\mathrm{field,obs}} > 0,$$

This excess is not treated as a presupposed particle halo term. It is treated as an empirical target whose internal structure can be tested directly.

Using galaxy-separated GroupKFold cross-validation, the analysis tests whether the positive acceleration excess can be reconstructed from nonlocal baryonic organization, local disk structure, global scale variables, and a smooth saturation term. Even the unsaturated field response model reaches a predictive accuracy close to the empirical radial acceleration relation (RAR). In the high-quality $Q = 1$ sample, the saturation-limited extension reaches essentially RAR-level accuracy,

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.144258 \text{ dex}, \qquad \mathrm{RMSE}_{\mathrm{RAR}} = 0.144091 \text{ dex},$$

and in the expanded $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples it lies below the corresponding RAR error values.

Several controls support the empirical structure. A galaxy-shuffle null test with 200 realizations shows that no random galaxy-level reassignment of the target values reaches the performance of the real model. A galaxy-wise bootstrap with 300 realizations shows a stable sign structure: the nonlocal dynamical kernel remains positive, the local disk structure remains positive, and the saturation term remains negative. Generic machine-learning baselines such as Random Forest, Extra Trees, and HistGradientBoosting also fail to outperform the interpretable saturation-limited model, even though they are tested on the same features and with the same galaxy-separated validation logic.

The analysis is extended by a theoretical interpretation of gravitational lensing in terms of the Universal Quantum Foam Hypothesis (UQSH). Lensing convergence is read as the projection of organized field structure. Galaxies act as baryonic anchors of overlapping, scale-dependent field configurations with saturation. On this basis, the Bullet Cluster is examined as an external geometric stress test. The CANUCS κ map shows two dominant convergence components that are robustly separated from the Chandra X-ray gas. The local JWST/F277W analysis also shows that the western κ component does not have to be treated as an empty lensing peak, but lies within a broader near-infrared environment.

The results do not prove a fundamental UQSH field equation, and they do not replace external rotation-curve data or full dynamical cluster simulations. They do, however, define a concrete, reproducible, and falsifiable empirical structure: the positive acceleration excess in SPARC rotation curves is described competitively with the RAR by a nonlocal, scale-dependent, saturation-limited field response parametrization. The next decisive tests are independent rotation-curve data, WISE- and GALEX-based proxies for structure and stellar populations, renewed ML controls, and further colliding clusters with lensing, X-ray, and near-infrared data.

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1. Introduction

> "A space without physical properties appears hardly conceivable."
> — paraphrased after Albert Einstein

1.1. Aim of the Empirical and Interpretive Analysis

The present work investigates dark matter phenomenology within a simple, reproducible, and rigorously tested field-response description. The central approach is that the observed additional acceleration and lensing contribution is not treated from the outset as the effect of a separately assumed invisible particle substance, but as an empirically testable response of a continuous field medium to baryonic organization, nonlocal structure, and saturation-limited redistribution (45).

This does not deny the dark matter effect. On the contrary, the observed excess is treated as a real dynamical target quantity. The new question is whether its structure can also be described as an organized field response without assuming a particle halo in advance.

The analysis has two distinct but related aims. First, the dynamical excess is determined reproducibly from the SPARC `rotmod` data (1). This step is purely empirical. From the observed rotation velocities and the baryonic contributions of gas, disk, and bulge, the pointwise excess

$$V^2_{\mathrm{excess}}(R) = V^2_{\mathrm{obs}}(R) - V^2_{\mathrm{bar}}(R)$$

is determined. Positive values are treated as a positive dynamical excess. Local negative values are not interpreted as negative dark matter, but are recorded separately as overshoot or modelling regions.

Second, the analysis tests whether the reconstructed structure of this excess is compatible with the UQSH interpretation. What matters is not merely the existence of the known effect, but its organization: its prevalence in the sample, its radial increase, and its coupling to visible baryonic structural quantities. If these features occur systematically, UQSH can interpret them as indications of a field-mechanical response of the medium, without immediately excluding particle-based dark matter.

The analysis therefore does not claim to rediscover the dark matter effect. It assumes its observed dynamical and geometrical character and asks what structure this phenomenology has. The central aim of this work is to test a concrete, reproducible, and falsifiable field-response parametrization that describes the positive acceleration excess in SPARC rotation curves competitively with the RAR and remains compatible with MDAR, BTFR, inner rotation-curve structure, and gravitational-lensing geometry. In addition, the UQSH interpretation of gravitational lensing as projected field organization is formulated and tested against the Bullet Cluster as an external stress test.

The theory was the origin of the approach, but not a substitute for the empirical test. The present work therefore does not test the entire UQSH ontology, but the concrete and falsifiable field-response structure motivated by it. The robustness of this structure is measured against reproducible observational data, galaxy-separated validation, null tests, bootstrap stability, ML comparisons, and an external Bullet-Cluster stress test.

The presentation focuses on the final reproducible form of the analysis. Earlier intermediate models and technical development steps are not treated as separate result sections. They are mentioned only where they are needed for methodological context. This keeps the main text focused on the tested saturation-limited field response, its statistical controls, its consistency tests, and its falsifiable

consequences.

2. Empirical Results

The empirical analysis is organized as a step-by-step test program. First, the data basis is defined and the observed acceleration is decomposed into a baryonic contribution and a positive acceleration excess. The empirical RAR is then determined as a strong reference relation. On this basis, the analysis tests whether the positive excess can be reconstructed directly from nonlocal baryonic organization, local disk structure, global scale, and negative saturation.

This is not just a test of an alternative curve fit. The question is whether the observed excess has a stable internal structure: nonlocal organization as an enhancing contribution, local disk structure as an additional contribution, and saturation as a limiting component.

The order of the results is:

1. data basis and definition of the positive acceleration excess,
2. empirical RAR as a compact reference relation,
3. direct field excess model with a nonlocal dynamical kernel,
4. saturation-limited extension of the field response model,
5. galaxy-shuffle null test against random galaxy assignment,
6. empirical main formula of the saturation-limited model,
7. bootstrap test of coefficient stability,
8. comparison with generic machine-learning baselines,
9. synthesis of the empirical results.

This structure deliberately separates empirical predictive performance, statistical control, and physical interpretation. In particular, the field-response interpretation is not inferred from fit quality alone. It is discussed only after testing for data leakage, shuffle robustness, coefficient stability, and comparison with generic regressors.

The following sections therefore first present the empirical target quantities, comparison models, and control tests. The field-response interpretation then builds on this reproducible result structure. It is not placed before the data, but developed from the tested structure of the positive acceleration excess.

2.1. Data Basis and Target Quantities

The empirical analysis is based on rotation-curve data from the SPARC data set. In total, 175 galaxies with 3391 radial data points were processed. For each radial point, the observed rotation velocity $V_{\mathrm{obs}}(r)$, the gas contribution $V_{\mathrm{gas}}(r)$, the disk contribution $V_{\mathrm{disk}}(r)$, and, where present, the bulge contribution $V_{\mathrm{bulge}}(r)$ are available.

The baryonic rotation contribution was constructed from the individual components as

$$V_{\mathrm{bar}}^2(r) = V_{\mathrm{gas}}(r)\,|V_{\mathrm{gas}}(r)| + \Upsilon_{\mathrm{disk}} V_{\mathrm{disk}}^2(r) + \Upsilon_{\mathrm{bulge}} V_{\mathrm{bulge}}^2(r),$$

with the mass-to-light factors

$$\Upsilon_{\mathrm{disk}} = 0.5, \qquad \Upsilon_{\mathrm{bulge}} = 0.7.$$

The gas contribution was treated with its sign because the SPARC rotmod files allow negative gas velocities as effective contributions.

All dynamical quantities are subsequently evaluated as accelerations. The velocity contributions provided by SPARC are therefore not identified directly with accelerations, but are converted into radial accelerations via $g = V^2/r$. Only after this conversion is the positive acceleration excess $g_{\mathrm{obs}} - g_{\mathrm{bar}}$ formed.

The radial accelerations were computed from the rotation velocities as

$$g_{\mathrm{obs}}(r) = \frac{V_{\mathrm{obs}}^2(r)}{r},$$

and

$$g_{\mathrm{bar}}(r) = \frac{V_{\mathrm{bar}}^2(r)}{r},$$

with all quantities converted into SI units. After the validity filters were applied, 3389 valid acceleration points remained.

The central target quantity of this work is the positive acceleration excess after subtracting the baryonic rotation contribution:

$$g_{\mathrm{field,obs}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r), \qquad g_{\mathrm{field,obs}}(r) > 0.$$

This excess is the empirical core structure of the dark matter effect in rotation curves. At this point, the index field does not denote an assumed field ontology. It denotes the working hypothesis that the remaining dynamical contribution can be modelled as a structured field response. In the standard picture, the same excess is usually attributed to a dark matter component. The present analysis tests instead whether it can be described reproducibly by nonlocal baryonic organization, local disk structure, global scale quantities, and negative saturation.

For logarithmic modelling, the target variable was therefore

$$\log_{10} g_{\mathrm{field,obs}}(r).$$

Points with nonpositive or nonfinite field excess were excluded from this part of the analysis. The resulting samples were considered separately by data quality:

$$Q = 1, \qquad Q = 1 + 2, \qquad Q = 1 + 2 + 3.$$

This avoids mixing high-quality and less certain rotation curves without control. The $Q = 1$ sample

serves as the primary high-quality analysis basis, while $Q = 1 + 2$ and $Q = 1 + 2 + 3$ are used as robustness and expanded samples.

In addition to radial quantities, galaxy-wide quantities from the SPARC metadata table were used, especially disk scale length R_d, baryonic mass M_{bar}, gas mass M_{gas}, gas fraction, surface brightness, and quality class Q. These quantities were not used to split individual radial points of the same galaxy between training and test sets. All predictive tests with train-test splitting were performed galaxy-separated.

2.2. Empirical RAR as Reference Model

The empirical radial acceleration relation was first used as the reference model. It describes the observed acceleration g_{obs} as a function of the baryonic acceleration g_{bar} through a compact one-parameter form.

The parameter $g_\dagger$ denotes the characteristic acceleration scale of the empirical radial acceleration relation. In the present analysis, it was determined from the valid SPARC acceleration points as

$$g_\dagger = 1.3357362308255698 \times 10^{-10}\ \mathrm{m\,s^{-2}}.$$

With this definition, the RAR reference form used here is

$$g_{\mathrm{RAR}}(g_{\mathrm{bar}}) = \frac{g_{\mathrm{bar}}}{1 - \exp\left(-\sqrt{g_{\mathrm{bar}}/g_\dagger}\right)}.$$

This corresponds to

$$\log_{10} g_\dagger = -9.8742792939613.$$

For the high-quality $Q = 1$ sample, the empirical RAR reached a predictive accuracy of

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.144091\ \mathrm{dex}.$$

For the expanded samples, the values were

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.163458\ \mathrm{dex} \qquad (Q = 1 + 2),$$

and

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.176583\ \mathrm{dex} \qquad (Q = 1 + 2 + 3).$$

The RAR thus constitutes a very robust, compact reference relation, whose individual galaxy fit and scatter structure have already been extensively investigated in earlier SPARC-based studies (2; 3; 19). Its importance lies in its high predictive power with very few parameters. At the same time, the RAR remains a central testing ground in recent work between MOND-like interpretations, ΛCDM, and empirical rotation-curve models (13; 14). For this reason, it is not treated in the present work as a countermodel to be refuted, but as a methodological minimum standard. An alternative field

response model is empirically relevant only if it reaches comparable predictive performance under galaxy-separated validation and at the same time provides additional interpretable structure.

It is important that the RAR itself does not explicitly model the positive field excess

$$g_{\mathrm{field,obs}} = g_{\mathrm{obs}} - g_{\mathrm{bar}}.$$

It directly describes the relation between g_{bar} and g_{obs}. The field response model studied here instead starts from the positive excess and asks whether this excess can be reproduced from baryonic organization, local structure, and saturation.

The RAR therefore serves in this analysis as an empirical reference, not as a model to be disproved. The success of the field response model is not measured by whether it formally replaces the RAR, but by whether it can reconstruct the same empirical relation from a more explicit and interpretable field-excess structure.

2.3. Direct Field Excess Model

After the empirical RAR was fixed as the reference, the positive field excess was modelled directly. The target quantity is

$$g_{\mathrm{field,obs}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r), \qquad g_{\mathrm{field,obs}}(r) > 0.$$

For the regression, the logarithmic target

$$y(r) = \log_{10} g_{\mathrm{field,obs}}(r)$$

was used. The central difference from the empirical radial acceleration relation is that the total acceleration g_{obs} is not fitted directly as a function of g_{bar} (2; 3). Instead, the model targets the positive excess contribution that remains after subtracting the baryonic rotation contribution. In this way, the additional acceleration is not introduced as an assumed halo term, but treated as an empirical target quantity.

The primary input variable was a nonlocal dynamical baryonic kernel F_{dyn}. This kernel describes a galaxy-wide organization quantity that includes not only the local radial point, but the baryonic distribution of the entire rotation curve. The test is whether the positive excess contains a nonlocal baryonic order structure that would not be visible in purely local point-by-point fits.

The unsaturated field response model had the schematic form

$$\log_{10} g_{\mathrm{field}} = a + \alpha \log_{10} F_{\mathrm{dyn}} + \beta \log_{10}\left(\frac{r}{R_d}\right) + \gamma \log_{10} \Sigma_{\mathrm{disk,local}}.$$

Here R_d denotes the disk scale length and $\Sigma_{\mathrm{disk,local}}$ a local disk structure quantity. The nonlocal kernel describes the organized baryonic distribution of the whole galaxy, while r/R_d and $\Sigma_{\mathrm{disk,local}}$ account for the radial position and local disk structure of each data point.

Validation was performed with galaxy-separated cross-validation. All radial points of a test galaxy were held out together. This prevents the model from seeing radial points of the same galaxy in training and testing at the same time.

For the high-quality $Q = 1$ sample, the unsaturated field response model reached an out-of-fold accuracy of

$$\mathrm{RMSE}_{\mathrm{field,OOF}} = 0.146829 \text{ dex}$$

for the reconstructed total acceleration. On the same data, the empirical RAR reached

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.144091 \text{ dex}.$$

The relative difference was only

$$1.90\%.$$

For the expanded samples, the values were

$$\mathrm{RMSE}_{\mathrm{field,OOF}} = 0.163603 \text{ dex} \qquad (Q = 1 + 2),$$

compared with

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.163458 \text{ dex},$$

and

$$\mathrm{RMSE}_{\mathrm{field,OOF}} = 0.176976 \text{ dex} \qquad (Q = 1 + 2 + 3),$$

compared with

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.176583 \text{ dex}.$$

Even the unsaturated field response model therefore reached almost the level of the empirical RAR. This is a central finding because the model does not enforce the RAR form directly. It reconstructs the positive field excess from nonlocal baryonic organization and local disk structure. The positive excess therefore does not appear as an arbitrary residual, but as a structured and predictable dynamical quantity.

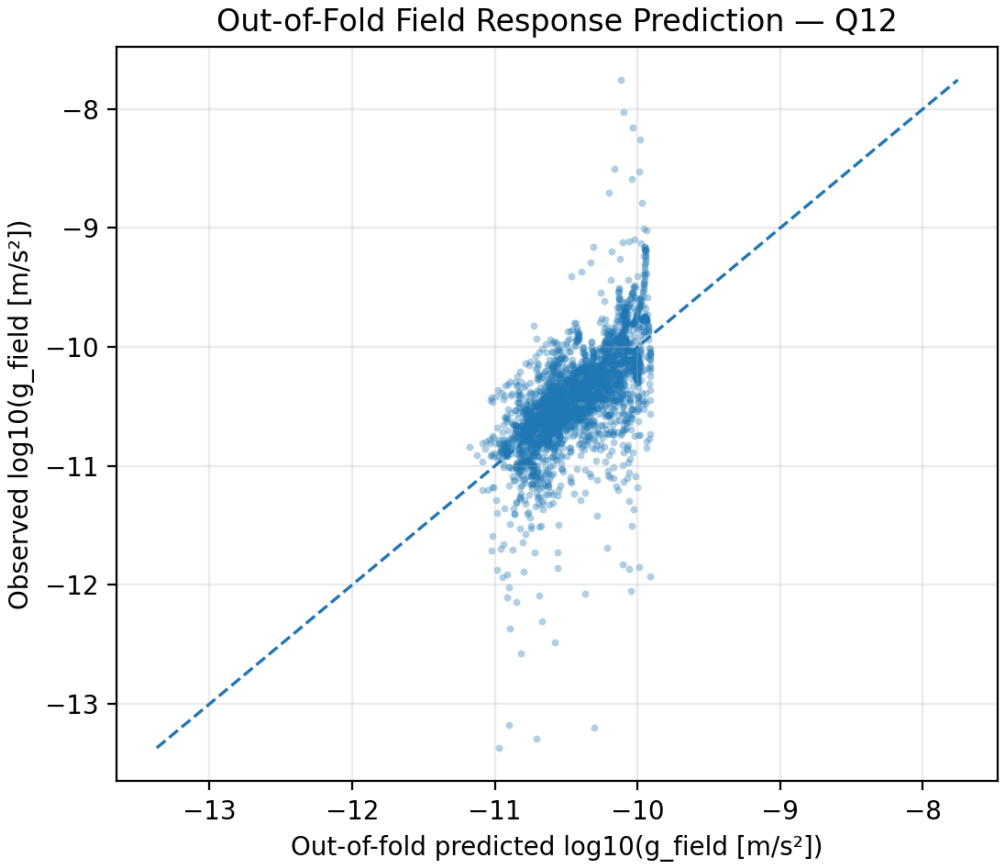


Figure 1: Galaxy-wise validated out-of-fold prediction of the positive acceleration excess for the Q=1+2 sample. The x-axis shows the predicted value of $\log_{10} g_{\mathrm{field}}$, and the y-axis shows the observed value. The dashed line marks the one-to-one relation. The point cloud shows that the positive acceleration excess is not randomly distributed, but can be predicted with reproducible structure from baryonic organization variables.

The out-of-fold structure does not show that the positive acceleration excess is new as such. This excess has long been known as the mass discrepancy, or as the difference between the observed acceleration and the acceleration expected from the baryonic component. In the RAR/MDAR literature, it is already captured implicitly through the relation between baryonic and observed acceleration. What is new here is the operationalization: the positive component $g_{\mathrm{obs}} - g_{\mathrm{bar}} > 0$ is modeled as an empirical target in its own right and tested, under galaxy-separated validation, for reproducible structure.

In this sense, the excess is not treated merely as an a posteriori difference relative to the RAR, but as a positively defined field response whose dependence on baryonic organization, local scale, and macroscopic saturation can be tested directly.

2.4. Saturation-Limited Field Response

The analysis of the unsaturated field excess model showed that nonlocal baryonic organization already reconstructs a large part of the positive field excess. At the same time, the residuals still showed systematic dependencies on global galaxy properties. This indicates that the field response cannot be described by a simple power-law enhancement alone, but needs a scale-dependent or saturation-limited

component.

The field response model was therefore extended by global scale terms and a smooth saturation term. This step tests whether the field response flattens at high effective organization strength instead of continuing to grow without limit as a power law. This reading connects to the previously formulated continuous field dynamics, where saturation is understood as a transition from further local enhancement to redistribution, counter-coupling, or a change of organization regime (45). The general form is

$$\log_{10} g_{\mathrm{field}} = a + \alpha \log_{10} F_{\mathrm{dyn}} + \beta \log_{10} \left(\frac{r}{R_d} \right) + \gamma \log_{10} \Sigma_{\mathrm{disk,local}} + \mu \log_{10} M_{\mathrm{scale}} + \nu \log_{10} R_d + \kappa S_{\mathrm{soft}}(X).$$

The saturation term was defined as a smooth flattening function:

$$S_{\mathrm{soft}}(X) = \log_{10} \left[1 + 10^{\log_{10} X - \log_{10} X_{\mathrm{sat}}} \right].$$

Here X_{sat} denotes the characteristic scale at which the effective organization strength enters the saturation regime. The decisive point is the sign of the coefficient. A negative coefficient,

$$\kappa < 0,$$

means that the saturation term flattens the field response at high organization strength. It then acts not as an additional free enhancer, but as a limit on the response.

For the high-quality $Q = 1$ sample, the saturation-limited model gave

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.144258 \text{ dex},$$

compared with

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.144091 \text{ dex}.$$

The difference from the RAR is therefore only

$$0.000166 \text{ dex},$$

corresponding to about

$$0.115\%.$$

Thus, in the high-quality sample, the saturation-limited field response model lies essentially at the level of the empirical RAR.

In the expanded samples, the result improves relative to the RAR. For $Q = 1 + 2$, the model reached

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.158329 \text{ dex},$$

while the empirical RAR gave

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.163458 \text{ dex}.$$

For $Q = 1 + 2 + 3$, the saturation-limited model reached

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.169357 \text{ dex},$$

compared with

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.176583 \text{ dex}.$$

The improvement over the unsaturated field response model was visible in all three samples. For $Q = 1$, the out-of-fold RMSE decreased from

$$0.146829$$

to

$$0.144258.$$

For $Q = 1 + 2$, it decreased from

$$0.163603$$

to

$$0.158329,$$

and for $Q = 1 + 2 + 3$ from

$$0.176976$$

to

$$0.169357.$$

The saturation term therefore improves not only a single sample, but acts consistently across different quality selections.

Table 1: Galaxy-separated validation comparison of the empirical RAR, the direct field-excess model, and the saturation-limited field-response model. RMSE values are given in dex. Negative values of $\Delta_{\mathrm{sat-RAR}}$ indicate an improvement relative to the RAR.

Sample	RAR RMSE	Direct field model	Saturation model	$\Delta_{\mathrm{sat-RAR}}$
Q=1	0.144091	0.146829	0.144258	0.000166
Q=1+2	0.163458	0.163603	0.158329	-0.005129
Q=1+2+3	0.176583	0.176976	0.169357	-0.007225

Table 1 shows that the saturation-limited extension improves the predictive performance of the direct field model in all quality selections. In the primary $Q = 1$ sample, it reaches practically the RAR level; in the extended $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, it lies below the corresponding RAR RMSE values. The decisive point is not only the numerical improvement, but that it arises from a structured negative saturation term and not from an arbitrarily flexible fit.

The extended quality samples are not treated as replacements for the high-quality $Q = 1$ analysis. The $Q = 1$ sample remains the primary reference. The $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples serve as robustness checks against less certain rotation curves. The finding is therefore not inferred from the mere inclusion of lower-quality data, but from the fact that the same sign structure and the negative saturation term are preserved across different quality selections.

The negative saturation term is therefore a central empirical result of this analysis. It shows that an unbounded power-law increase of the field response describes the data less well than a form in which the organizational effect flattens at high effective strength. Saturation is therefore not merely an additional fit parameter, but the key structure through which the field-response model changes from a pure amplification form into a bounded response form.

The empirical finding imposes a clear theoretical minimum requirement: any UQSH-like field description of the positive acceleration excess must contain a finite scale, structure, or self-limitation of the response.

2.5. Galaxy-Shuffle Null Test

To test whether the saturation-limited field response model merely exploits a random target-value structure, a galaxy-wise shuffle null test was performed. The internal radial structure of each galaxy was kept intact, while the target values of the positive field excess were reassigned between galaxies.

This test is stricter than pointwise shuffling of individual radial points. Pointwise shuffling would destroy the physical structure inside a rotation curve. The galaxy-shuffle used here preserves the radial coherence of individual galaxies, but breaks the real assignment between a galaxy structure and its field excess. This tests whether the real galaxy assignment contains predictive information.

For each sample, 200 shuffle realizations were computed. The real saturation-limited model was then compared with the distribution of the shuffle models.

For the high-quality $Q = 1$ sample, the result was

$$\mathrm{RMSE}_{\mathrm{real}} = 0.307699,$$

while the shuffle models reached on average

$$\langle \mathrm{RMSE}_{\mathrm{shuffle}} \rangle = 0.376844.$$

The best shuffle realization had

$$\mathrm{RMSE}_{\mathrm{shuffle,min}} = 0.334426.$$

Even the best shuffle model was therefore worse than the real model.

For the expanded $Q = 1 + 2$ sample, the result was

$$\mathrm{RMSE}_{\mathrm{real}} = 0.327736,$$

compared with

$$\langle \mathrm{RMSE}_{\mathrm{shuffle}} \rangle = 0.439390.$$

The best shuffle realization had

$$\mathrm{RMSE}_{\mathrm{shuffle,min}} = 0.361046.$$

Here too, no shuffle model reached the performance of the real model.

For the full $Q = 1 + 2 + 3$ sample, the result was

$$\mathrm{RMSE}_{\mathrm{real}} = 0.350915,$$

compared with

$$\langle \mathrm{RMSE}_{\mathrm{shuffle}} \rangle = 0.487167.$$

The best shuffle realization had

$$\mathrm{RMSE}_{\mathrm{shuffle,min}} = 0.420344.$$

In all three samples,

$$f_{\mathrm{shuffle} \leq \mathrm{real}} = 0.$$

With 200 shuffle realizations, this corresponds to the empirical bound

$$p < \frac{1}{201} \simeq 0.004975.$$

The null test therefore shows that the real assignment between baryonic galaxy structure and positive acceleration excess contains predictive information. The model does not merely use an arbitrary global point-cloud structure. It uses a real galaxy-level order. If the model performance arose only from the

general target-value distribution or from pointwise correlations, some shuffle realizations would have to reach or outperform the real assignment. This is not the case in the 200 tested realizations.

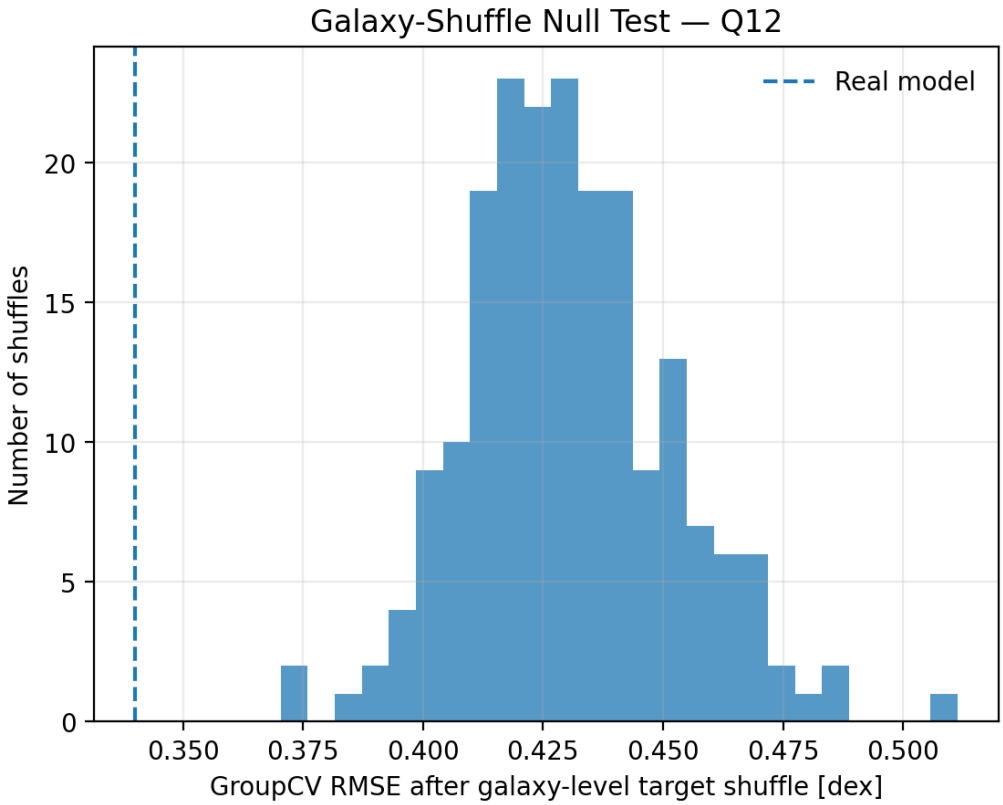


Figure 2: Galaxy-shuffle null test for the Q=1+2 sample. The histogram shows the GroupCV RMSE values after random reassignment of the galaxy-level target values. Smaller values indicate better predictive performance. The dashed line marks the real model. None of the 200 shuffle realizations reaches or falls below the error of the real assignment. This shows that the actual galactic structure and target assignment carry predictive information.

The null test complements pure predictive accuracy with an assignment test at the galaxy level. The important point is not only that the model reconstructs the positive acceleration excess, but that this reconstruction is tied to the real combination of baryonic galaxy structure and target values. If the model performance were mainly explained by the global target-value distribution or by arbitrary pointwise correlations, some shuffle realizations would have to reach or outperform the real assignment. This is not the case in the 200 tested realizations.

The shuffle test does not provide an independent field ontology. It does, however, provide a hard empirical boundary condition: any successful interpretation of the positive acceleration excess must explain why the real galaxy-wise baryonic organization carries predictive information, while galaxy-wise reassigned targets perform systematically worse.

2.6. Empirical Main Formula of the Saturation-Limited Model

The saturation-limited field response analysis yields an empirical working form for the positive acceleration excess. This formula is not meant as a fundamental field law, but as a data-driven

parametrization of the excess structure observed in SPARC. It connects to the previously formulated continuous field dynamics, in which nonlocal reorganization, scale dependence, and saturation appear as central structural features of the field response (45). The general form is

$$\log_{10} g_{\mathrm{field}} = a+\alpha \log_{10} F_{\mathrm{dyn}}+\beta \log_{10}\left(\frac{r}{R_d}\right)+\gamma \log_{10} \Sigma_{\mathrm{disk,local}}+\mu \log_{10} M_{\mathrm{scale}}+\nu \log_{10} R_d+\kappa S_{\mathrm{soft}}(X).$$

The saturation term is defined by

$$S_{\mathrm{soft}}(X) = \log_{10}\left[1 + 10^{\log_{10} X-\log_{10} X_{\mathrm{sat}}}\right].$$

For reproducibility, the fitted coefficients and selected structural quantities of the saturation-limited model are summarized in Table 2. The coefficients $a, \alpha, \beta, \gamma, \mu, \nu, \kappa$ were obtained by regression for each fixed model specification and therefore did not require iterative initial values. The quantities λ, M_{scale}, and $\log_{10} X_{\mathrm{sat}}$ specify the selected nonlocal kernel scale, the global mass proxy, and the effective saturation transition scale.

The values of $\log_{10} X_{\mathrm{sat}}$ are empirical transition values of the tested soft-saturation term. They result from the model selection within the SPARC analysis and are not interpreted here as fundamental constants. A later theoretical formulation must examine whether such effective saturation transitions are related to scale-dependent astrophysical thresholds or to more fundamental field-dynamical boundary conditions.

Table 2: Coefficients and fixed structural quantities of the saturation-limited field-response model for the three quality samples. For each fixed model specification, the linear coefficients were determined directly by regression and therefore required no iterative initial values. Dashes indicate terms that were not used in the respective specification. The negative coefficient of $S_{\mathrm{soft}}(X)$ is preserved in all samples and forms the empirical saturation term.

Term / fixed quantity	Q=1	Q=1+2	Q=1+2+3
a	-11.160545	-13.669981	-14.016465
$\log F_{\mathrm{dyn},\lambda=0.25}$	0.698163	–	–
$\log F_{\mathrm{dyn},\lambda=0.5}$	–	0.951818	1.083024
$\log M_{\mathrm{bar}}$	–	0.072058	0.081234
$\log M_{\mathrm{gas}}$	-0.135596	–	–
$\log R_d$	-0.241819	-0.570066	-0.591970
$\log \Sigma_{\mathrm{disk,local}}$	0.103436	0.079119	0.078984
$\log(r/R_d)$	-0.187087	-0.286101	-0.316931
$S_{\mathrm{soft}}(X)$	-0.368213	-0.649829	-0.743764
Kernel scale λ	0.25	0.5	0.5
Saturation transition $\log_{10} X_{\mathrm{sat}}$	2.8	2.8	2.6
Global mass proxy M_{scale}	M_{gas}	M_{bar}	M_{bar}

This explicitly gives both the regressively determined coefficients and the fixed structural quantities required for reproduction: the kernel scale λ, the effective saturation transition $\log_{10} X_{\mathrm{sat}}$, and the global mass proxy M_{scale}. The decisive information is not the exact numerical identity of each individual coefficient, but the stable sign structure: the dynamical organization term contributes positively, the local disk density also contributes positively, while the radial disk scaling acts negatively. Most importantly, the saturation term remains negative in all quality samples.

The empirically relevant condition is

$$\kappa < 0.$$

The saturation term therefore does not act as an additional enhancement, but as a limit on the field response at high effective organization strength.

For the high-quality $Q = 1$ sample, the primary working form is

$$\begin{aligned}\log_{10} g_{\mathrm{field}} = &- 11.160545 + 0.698163 \log_{10} F_{\mathrm{dyn},0.25} - 0.187087 \log_{10}\left(\frac{r}{R_d}\right) \\ &+ 0.103436 \log_{10} \Sigma_{\mathrm{disk,local}} - 0.135596 \log_{10} M_{\mathrm{gas}} - 0.241819 \log_{10} R_d \\ &- 0.368213\, S_{\mathrm{soft}}(X),\end{aligned}$$

with

$$\log_{10} X_{\mathrm{sat}} = 2.8.$$

For the expanded $Q = 1 + 2$ sample, the corresponding form is

$$\begin{aligned}\log_{10} g_{\mathrm{field}} = &- 13.669981 + 0.951818 \log_{10} F_{\mathrm{dyn},0.5} - 0.286101 \log_{10}\left(\frac{r}{R_d}\right) \\ &+ 0.079119 \log_{10} \Sigma_{\mathrm{disk,local}} + 0.072058 \log_{10} M_{\mathrm{bar}} - 0.570066 \log_{10} R_d \\ &- 0.649829\, S_{\mathrm{soft}}(X),\end{aligned}$$

with

$$\log_{10} X_{\mathrm{sat}} = 2.8.$$

For the full $Q = 1 + 2 + 3$ sample, the result is

$$\begin{aligned}\log_{10} g_{\mathrm{field}} = &- 14.016465 + 1.083024 \log_{10} F_{\mathrm{dyn},0.5} - 0.316931 \log_{10}\left(\frac{r}{R_d}\right) \\ &+ 0.078984 \log_{10} \Sigma_{\mathrm{disk,local}} + 0.081234 \log_{10} M_{\mathrm{bar}} - 0.591970 \log_{10} R_d \\ &- 0.743764\, S_{\mathrm{soft}}(X),\end{aligned}$$

with

$$\log_{10} X_{\mathrm{sat}} = 2.6.$$

The sign structure is consistent across all three quality selections. The nonlocal dynamical kernel has a positive coefficient. The local disk structure contributes positively to the field response. The saturation term has a negative coefficient and limits the response at high effective organization strength. Radius and scale terms act mainly as damping terms. The important point is therefore not fit quality alone, but the repeated structure of coefficient directions. It is the empirical main formula of the tested field response model: an interpretable approximation that describes how the positive acceleration excess

in SPARC rotation curves is reconstructed from nonlocal baryonic organization, local disk structure, global scale, and saturation. Its significance lies not in the individual numerical coefficient values, but in the stable sign structure, the reproducible predictive performance, and the falsifiability of this structure in independent data.

2.7. Coefficient Stability from Galaxy-Wise Bootstrap

In addition to predictive accuracy, the analysis tested whether the central sign structure of the saturation-limited field response model is stable. A galaxy-wise bootstrap test was used for this purpose. In each bootstrap realization, galaxies were resampled with replacement, while all radial points of each selected galaxy were taken together. The model was then fitted again.

This test does not only examine the scatter of individual parameters. It mainly tests the stability of the physically relevant sign directions. For the field response model, this is decisive. The most robust finding is not the exact coefficient value, but whether nonlocal organization, local disk structure, and saturation retain the same direction of action across resampled galaxy samples:

1. The nonlocal dynamical kernel should remain positive.
2. The local disk structure should contribute positively.
3. The saturation term should remain negative.
4. Radius and scale terms should act mainly as damping terms.

For each of the three samples, 300 bootstrap realizations were computed.

In the high-quality $Q = 1$ sample, the dynamical kernel gave

$$\langle \alpha \rangle = 0.711987, \qquad \sigma_\alpha = 0.166312,$$

with

$$P(\alpha > 0) = 1.000000.$$

The nonlocal dynamical kernel therefore remained positive in all bootstrap realizations.

For the saturation term, the result was

$$\langle \kappa \rangle = -0.406888, \qquad \sigma_\kappa = 0.187629,$$

with

$$P(\kappa < 0) = 0.993333.$$

The bootstrap test therefore confirms the central sign structure of the model. Nonlocal dynamical organization remains enhancing across the resampled galaxy samples, local disk structure also remains enhancing, and the saturation term remains limiting. The test shows that the most important directions of the empirical main formula do not depend on a single random galaxy selection. This stability is

what makes the structure physically interpretable, even though it still does not imply a fundamental field equation.

The local disk term also remained stably positive:

$$\langle \gamma \rangle = 0.108930, \qquad \sigma_\gamma = 0.038487,$$

with

$$P(\gamma > 0) = 1.000000.$$

In the $Q = 1$ sample, the radius term showed somewhat weaker, but still predominantly negative stability:

$$\langle \beta \rangle = -0.182185, \qquad \sigma_\beta = 0.107837,$$

with

$$P(\beta < 0) = 0.946667.$$

This value lies just below a strict 95 percent threshold, but the median remains clearly negative. The $Q = 1$ sample therefore shows a robust core structure, while the radius term is somewhat more sensitive to the galaxy selection.

In the expanded $Q = 1 + 2$ sample, the stability was even clearer. For the dynamical kernel,

$$P(\alpha > 0) = 1.000000,$$

for the saturation term,

$$P(\kappa < 0) = 1.000000,$$

for the radius term,

$$P(\beta < 0) = 1.000000,$$

and for the local disk structure,

$$P(\gamma > 0) = 1.000000.$$

In the full $Q = 1 + 2 + 3$ sample, the sign structure also remained stable:

$$P(\alpha > 0) = 1.000000, \qquad P(\kappa < 0) = 1.000000, \qquad P(\beta < 0) = 1.000000.$$

For the local disk term, the result was

$$P(\gamma > 0) = 0.996667.$$

The bootstrap test therefore confirms the central empirical structure of the model across all quality selections. Nonlocal dynamical organization acts as an enhancing term, local disk structure also acts as an enhancing term, and the saturation term acts as a limiting term. The same sign logic that appears in the main formula is retained under galaxy-wise resampling.

2.8. Comparison with Generic Machine-Learning Baselines

A central methodological objection to the saturation-limited field response model concerns its greater flexibility compared with the empirical RAR. In the comparison used here, the RAR is essentially described by one characteristic acceleration parameter, whereas the field response model contains several structure, scale, and saturation terms. For this reason, the analysis also tested whether generic machine-learning models using the same input variables clearly outperform the saturation-limited model. This comparison is not a complete competition against all possible regression methods. It is a control test against obvious flexible models with greater functional freedom.

Random Forest, Extra Trees, and HistGradientBoosting were compared (36; 37; 38). All models were tested on the same rows, the same features, and with the same galaxy-separated GroupKFold cross-validation. The comparison is therefore methodologically direct: no model was allowed to see radial points of the same galaxy in both training and testing.

For the high-quality $Q = 1$ sample, the empirical RAR reached

$$\mathrm{RMSE}_{\mathrm{RAR}} = 0.144091,$$

while the saturation-limited interpretable field response model reached

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.144258.$$

The generic ML models were clearly higher:

$$\mathrm{RMSE}_{\mathrm{ExtraTrees}} = 0.159545,$$

$$\mathrm{RMSE}_{\mathrm{RandomForest}} = 0.165255,$$

and

$$\mathrm{RMSE}_{\mathrm{HistGradientBoosting}} = 0.172536.$$

For the expanded $Q = 1 + 2$ sample, the saturation-limited field response model was even below the empirical RAR:

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.158329 < \mathrm{RMSE}_{\mathrm{RAR}} = 0.163458.$$

The ML baselines again remained weaker:

$$\mathrm{RMSE}_{\mathrm{ExtraTrees}} = 0.171279,$$

$$\mathrm{RMSE}_{\mathrm{RandomForest}} = 0.181008,$$

and

$$\mathrm{RMSE}_{\mathrm{HistGradientBoosting}} = 0.191570.$$

For the full $Q = 1 + 2 + 3$ sample, the same pattern appeared:

$$\mathrm{RMSE}_{\mathrm{sat}} = 0.169357 < \mathrm{RMSE}_{\mathrm{RAR}} = 0.176583,$$

while the ML models gave

$$\mathrm{RMSE}_{\mathrm{ExtraTrees}} = 0.182332,$$

$$\mathrm{RMSE}_{\mathrm{RandomForest}} = 0.193315,$$

and

$$\mathrm{RMSE}_{\mathrm{HistGradientBoosting}} = 0.199358.$$

Table 3: Comparison of the saturation-limited interpretable field response model with generic machine-learning baselines under galaxy-wise validation. The out-of-fold RMSE of the reconstructed observed acceleration g_{obs} is given in dex. Smaller values indicate better predictive performance.

Sample	Saturation model	Extra Trees	Random Forest	Hist. Grad. Boosting
Q=1	0.144258	0.159545	0.165255	0.172536
Q=1+2	0.158329	0.171279	0.181008	0.191570
Q=1+2+3	0.169357	0.182332	0.193315	0.199358

If the observed model performance were mainly the result of arbitrary flexible function approximation, models such as Random Forest, Extra Trees, or HistGradientBoosting should generalize at least comparably well on the same features and under the same galaxy-separated validation.

They do not. The smooth, interpretable saturation form remains better than the tested ML baselines in all three quality samples. This does not prove the field-response interpretation, but it weakens an important objection: the model performance does not obviously arise from maximal functional freedom. It arises from a restricted structure with nonlocal organization, scale coupling, and a negative saturation term.

The comparison with generic ML baselines does not replace an external test. It does, however, sharpen the methodological point: the smooth saturation structure is not obviously a weaker or arbitrary choice

compared with flexible data models. External data sets and additional observable proxies remain the next methodological step.

Taken together, the empirical results give a clear picture. Based on SPARC, the positive acceleration excess

$$g_{\mathrm{field,obs}} = g_{\mathrm{obs}} - g_{\mathrm{bar}}$$

is not merely a residual of the empirical radial acceleration relation (2; 3), but a target quantity that can be modelled in its own right. Even the unsaturated model with a nonlocal dynamical kernel reaches almost the level of this reference relation under galaxy-separated cross-validation. Baryonic organization therefore carries real predictive information about the acceleration excess.

The subsequent extension with global scale terms and a smooth saturation term improves the model performance consistently. In the high-quality $Q = 1$ sample, the saturation-limited field response model lies essentially at RAR level. In the expanded $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, it lies slightly below the RAR RMSE. The decisive point, however, is not only the numerical improvement, but the stable sign structure: the dynamical kernel remains positive, the local disk structure remains positive, and the saturation term remains negative. This structure was confirmed by galaxy-wise bootstrap.

The control tests strengthen this interpretation. The galaxy-shuffle null test shows that the real assignment between galaxy structure and positive field excess is not reproduced by random galaxy-wise target reassignment. The comparison with generic machine-learning baselines also shows that Random Forest, Extra Trees, and HistGradientBoosting do not outperform the saturation-limited model on the same features and under the same galaxy-separated validation. The empirical structure is therefore not simply explained by arbitrary flexible function approximation.

The SPARC analysis therefore yields a concrete empirical structure: a smooth, interpretable, saturation-limited field response parametrization that is competitive with the RAR, robust against shuffling, stable in its coefficients, and competitive against the tested generic ML baselines. What does not yet follow is a final fundamental UQSH field equation or a replacement for external validation on independent data.

The following consistency tests examine whether this structure exists not only in the direct modelling of the positive acceleration excess, but also survives in related representations of the rotation-curve data. These include the acceleration ratio in MDAR space, the global baryonic Tully-Fisher relation, and the inner structure of the rotation curves.

Although the saturation-limited field-response model contains more explicit structure terms than the one-parameter RAR reference, its additional freedom is constrained in three ways. First, all predictive tests are galaxy-separated, so no radial points of the same galaxy enter training and testing simultaneously. Second, the sign structure of the main coefficients is not chosen freely after the fact, but tested under galaxy-wise bootstrap resampling. Third, generic regressors with greater functional flexibility do not outperform the interpretable model on the same features and validation logic. The comparison is therefore not claimed as a parameter-economy victory over the RAR, but as evidence that the positive acceleration excess contains reproducible structure beyond a purely local acceleration relation.

As an additional model-complexity check, we computed a predictive information-criterion-like score from the galaxy-separated out-of-fold residuals,

$$\mathrm{IC}_{\mathrm{pred}} = n \ln\left(\mathrm{RMSE}^2_{\mathrm{OOF}}\right) + 2k.$$

This quantity is not used as a classical in-sample AIC, but as a CV-AIC-like score based on the same held-out galaxy predictions used for the RMSE comparison. The parameter counts were taken as $k_{\mathrm{RAR}} = 1$, $k_{\mathrm{direct}} = 4$, and $k_{\mathrm{sat}} = 7$. As a conservative check, the saturation-limited model was also evaluated with $k_{\mathrm{sat}} = 10$, counting the selected kernel scale, mass proxy, and saturation transition as additional model-selection degrees of freedom.

Table 4: Predictive information-criterion-like comparison based on galaxy-separated out-of-fold residuals. The score is defined as $\mathrm{IC}_{\mathrm{pred}} = n \ln(\mathrm{RMSE}^2_{\mathrm{OOF}}) + 2k$. It is not a classical in-sample AIC, but a CV-AIC-like model-complexity check. Negative $\Delta\mathrm{IC}_{\mathrm{pred}}$ values indicate preference over the RAR reference.

Sample	Model	n	k	RMSE	$\Delta\mathrm{IC}_{\mathrm{pred}}$
$Q = 1$	RAR	5577	1	0.144091	0.0
$Q = 1$	Direct field model	5577	4	0.146829	+216.0
$Q = 1$	Saturation-limited model	5577	7	0.144258	+24.9
$Q = 1$	Saturation model, conservative	5577	10	0.144258	+30.9
$Q = 1 + 2$	RAR	7469	1	0.163458	0.0
$Q = 1 + 2$	Direct field model	7469	4	0.163603	+19.2
$Q = 1 + 2$	Saturation-limited model	7469	7	0.158329	-464.2
$Q = 1 + 2$	Saturation model, conservative	7469	10	0.158329	-458.2
$Q = 1 + 2 + 3$	RAR	7560	1	0.176583	0.0
$Q = 1 + 2 + 3$	Direct field model	7560	4	0.176976	+39.6
$Q = 1 + 2 + 3$	Saturation-limited model	7560	7	0.169357	-619.7
$Q = 1 + 2 + 3$	Saturation model, conservative	7560	10	0.169357	-613.7

In the strict $Q = 1$ sample, the RAR remains favored by this criterion, as expected from its nearly identical RMSE and smaller parameter count. In the expanded $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, however, the saturation-limited model is strongly favored even under the conservative parameter count. The corresponding differences relative to the RAR are $\Delta\mathrm{IC}_{\mathrm{pred}} = -458.2$ for $Q = 1 + 2$ and $\Delta\mathrm{IC}_{\mathrm{pred}} = -613.7$ for $Q = 1 + 2 + 3$. The unsaturated direct field model is not favored by this criterion. This supports the interpretation that the empirical gain comes specifically from the saturation-limited structure rather than from a generic increase in model complexity.

3. Further Consistency Tests

The following tests examine whether the saturation-limited field response model also holds in related representations of the rotation-curve data. MDAR, BTFR, and inner mass-slope structures are not fully independent data sets, since they are derived from the same or closely related SPARC quantities. They are nevertheless necessary consistency checks. A viable field response model must not only reconstruct the pointwise acceleration g_{obs}, but must also remain compatible with the acceleration ratio, the global baryonic scaling relation, and the inner structure of the rotation curves.

These tests therefore do not expand the data basis. Instead, they sharpen the internal requirement on the model. A mere point-by-point fit would be insufficient if it generated contradictory structures in MDAR space, in the BTFR residual structure, or in the inner organization of rotation curves.

3.1. MDAR Consistency Test

In addition to the direct prediction of $g_{\rm obs}$, the saturation-limited field-response model was also tested in the MDAR/RAR space (2; 3). The relevant quantity is the dynamical acceleration ratio

$$D_{\rm MDAR} = \frac{g_{\rm obs}}{g_{\rm bar}},$$

together with the corresponding model and RAR quantities

$$D_{\rm sat} = \frac{g_{\rm obs,sat}}{g_{\rm bar}}, \qquad D_{\rm RAR} = \frac{g_{\rm RAR}}{g_{\rm bar}}.$$

The evaluation was performed in logarithmic space, $\log_{10} D_{\rm MDAR}$. Since this representation is mathematically closely connected to the direct acceleration prediction, the MDAR test does not provide an independent external data set. It does, however, tighten the internal consistency requirement: the model must not only reconstruct $g_{\rm obs}$, but also preserve the mass-discrepancy representation $g_{\rm obs}/g_{\rm bar}$ without producing a contradictory structure.

This is indeed the case. In the high-quality $Q = 1$ sample, the saturation-limited model remains essentially at the level of the empirical RAR also in MDAR space. In the extended $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, the slight improvement relative to the RAR found in the direct acceleration analysis is retained.

The MDAR test therefore confirms the previously found structure in a different representation: the saturation-limited field response not only reconstructs the observed total acceleration, but also remains consistent with the dynamical ratio $g_{\rm obs}/g_{\rm bar}$.

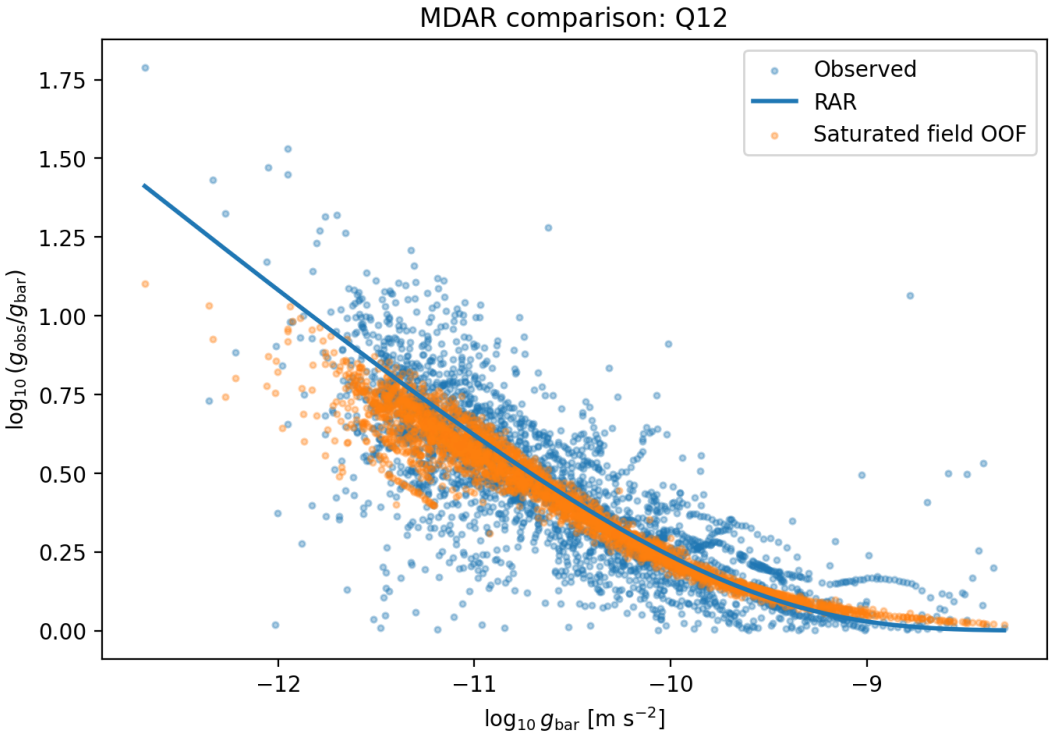


Figure 3: MDAR consistency test for the Q=1+2 sample. The logarithmic acceleration ratio $\log_{10}(g_{\rm obs}/g_{\rm bar})$ is shown as a function of $\log_{10} g_{\rm bar}$. The observed SPARC points are compared with the empirical RAR and the galaxy-separated out-of-fold prediction of the saturation-limited field-response model. The figure shows that the saturation-limited model reproduces the mass-discrepancy structure in MDAR space and preserves the RAR-like structure in this representation.

3.2. BTFR Consistency Test

In addition to the pointwise acceleration structure, the baryonic Tully-Fisher relation (BTFR) was examined as a galaxy-wide consistency relation (5). While RAR and MDAR compare radial acceleration points, the BTFR describes a global relation between baryonic mass and asymptotic rotation velocity:

$$\log_{10} M_{\rm bar} = A + B \log_{10} V_{\rm flat}.$$

For the high-quality $Q = 1$ sample, the result was

$$A = 2.585720, \qquad B = 3.557839,$$

with

$$\mathrm{RMSE}_{\rm BTFR} = 0.226093 \text{ dex}, \qquad R^2 = 0.924208.$$

For the expanded $Q = 1 + 2$ sample, the result was

$$A = 2.916817, \qquad B = 3.413261,$$

with

$$\mathrm{RMSE}_{\mathrm{BTFR}} = 0.247747 \text{ dex}, \qquad R^2 = 0.910985.$$

For the full $Q = 1 + 2 + 3$ sample, the result was

$$A = 3.399839, \qquad B = 3.189205,$$

with

$$\mathrm{RMSE}_{\mathrm{BTFR}} = 0.288828 \text{ dex}, \qquad R^2 = 0.881624.$$

The increasing scatter and the shallower slope when lower-quality classes are included are consistent with the expectation that the high-quality $Q = 1$ sample provides the most stable global scaling relation.

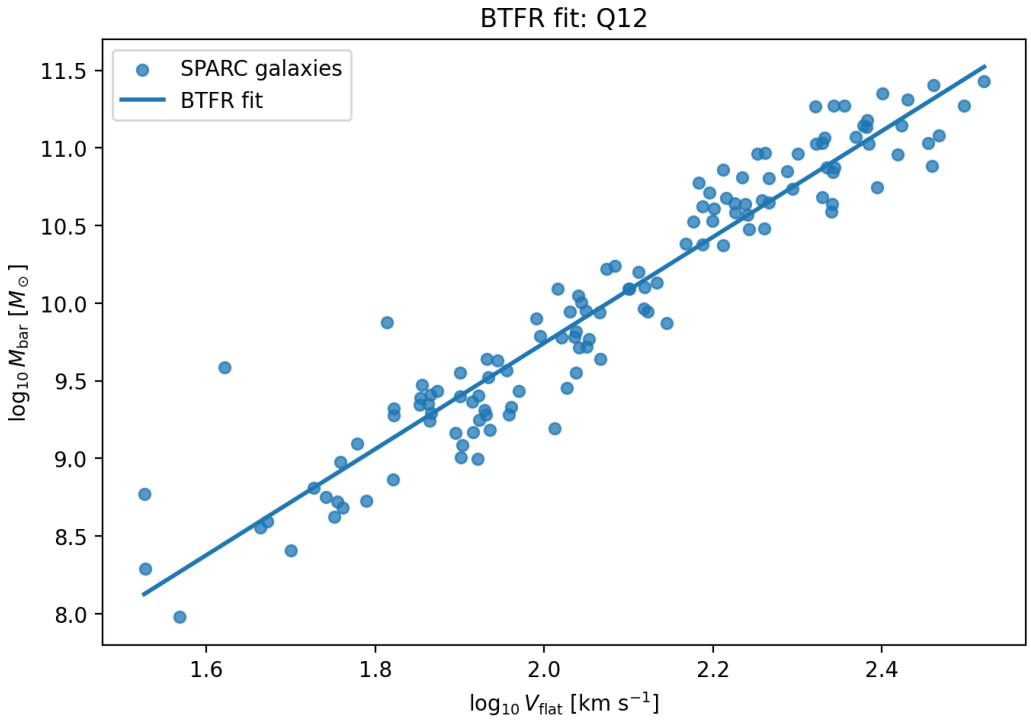


Figure 4: Baryonic Tully-Fisher relation for the Q=1+2 sample. The global relation between baryonic mass and flat rotation velocity is shown. The figure serves as an internal consistency check: the saturation-limited field response model must not only reproduce pointwise acceleration structures, but must also remain compatible with the global baryonic mass-velocity scaling.

Of particular interest is the relation between BTFR residuals and the galaxy-wise residuals of the saturation-limited field response model. The BTFR residuals were compared with the median residuals of the field response model at the galaxy level. The analysis was performed sample-consistently, so

that each galaxy was included only once in the respective quality sample.

For $Q = 1$, the Spearman correlation between the BTFR residual and the median field response residual was

$$r_{\mathrm{Spearman}} = -0.621856, \qquad p = 1.29 \times 10^{-10}.$$

For $Q = 1 + 2$, the result was

$$r_{\mathrm{Spearman}} = -0.702829, \qquad p = 1.64 \times 10^{-20}.$$

For $Q = 1 + 2 + 3$, the result was

$$r_{\mathrm{Spearman}} = -0.709687, \qquad p = 5.68 \times 10^{-22}.$$

The BTFR scatter is therefore systematically coupled to the residual structure of the field response. Galaxies that deviate from the global BTFR also show systematic deviations in the radial modelling of the field excess. In this analysis, the global mass-velocity scaling and the radial excess structure are therefore not independent phenomena, but two coupled representations of the same galactic dynamics.

This finding is an internal consistency check. It does not replace an independent data set, but it sharpens the requirement on the model. A viable field response description must not only be compatible with the RAR or MDAR point by point, but must also remain compatible with the global baryonic scaling relation. This coupling between global BTFR structure and radial field response residual structure is observed in all three quality selections.

3.3. Inner Mass Slope and Core-Cusp-Related Structure Test

After the pointwise field-excess model, an additional test was performed to determine whether the inner structure of the rotation curves is systematically coupled to global galaxy organization and field-response quantities. This test should be understood as a core-cusp-related structure test, not as a direct reconstruction of a dark-matter density profile.

From the observed circular dynamics, an effective enclosed mass quantity was derived,

$$M_{\mathrm{eff}}(r) = \frac{g_{\mathrm{obs}}(r) r^2}{G}.$$

This quantity does not describe a directly reconstructed density distribution, but a rotation-curve-based structural measure. In the inner region, a power-law form

$$M_{\mathrm{eff}}(r) \sim r^{\alpha}$$

was fitted, so that

$$\alpha = \frac{d \log M_{\mathrm{eff}}}{d \log r}$$

describes the effective inner mass increase. A value $\alpha \simeq 2$ corresponds approximately to NFW-like

inner behavior, whereas $\alpha \simeq 3$ corresponds to a constant-density inner region. Values below $\alpha \simeq 2$ indicate an even lower effective inner mass increase and should therefore not be identified uncritically with classical constant-density cores (6; 7; 15).

To test robustness, α was determined from the innermost $n = 3$, $n = 5$, and $n = 8$ rotation-curve points. The main analysis uses $n = 5$ inner points. In total, all 175 SPARC galaxies entered this reproducible reconstruction.

The main analysis shows a clear coupling of the inner mass slope to the global system size. For the correlation between α and the maximum rotation-curve radius $r_{\max}$, the results are

$$r_{\mathrm{Spearman}} = -0.434000, \qquad p = 7.19 \times 10^{-6} \quad (Q = 1),$$

$$r_{\mathrm{Spearman}} = -0.443176, \qquad p = 3.14 \times 10^{-9} \quad (Q = 1 + 2),$$

and

$$r_{\mathrm{Spearman}} = -0.406568, \qquad p = 2.37 \times 10^{-8} \quad (Q = 1 + 2 + 3).$$

Larger systems therefore show, on average, smaller values of α. The inner rotation-curve structure thus does not appear as a purely local detail, but as a scale-dependent property of the galaxy as a whole.

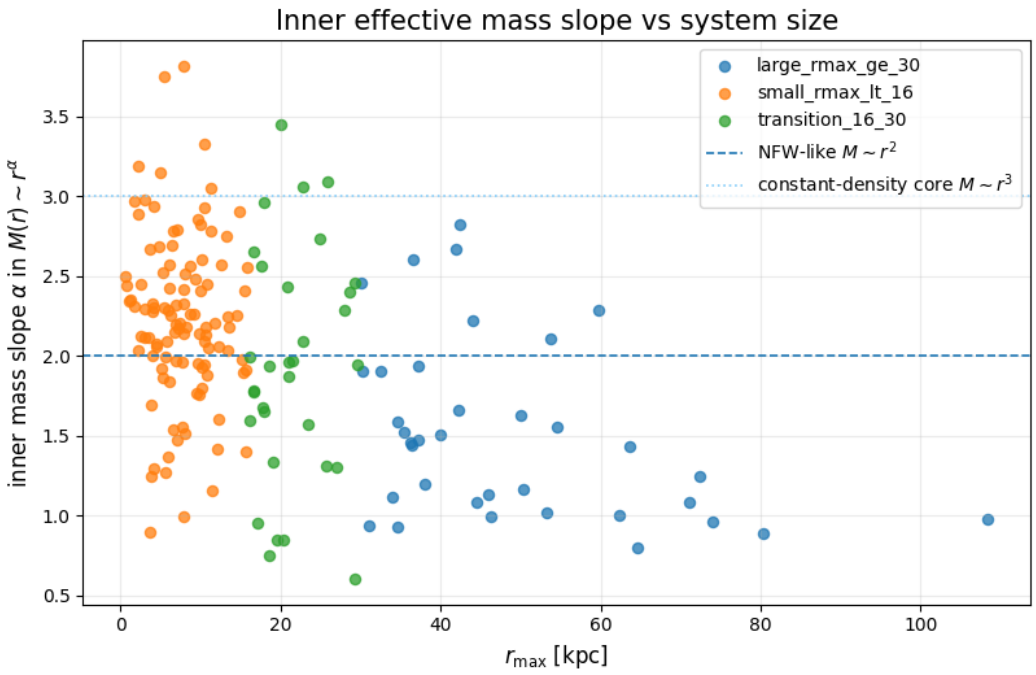


Figure 5: Core-cusp-related structure test using the inner effective mass slope α, determined from the five innermost rotation-curve points. The figure shows the relation between α and the maximum rotation-curve radius $r_{\max}$. It demonstrates that the inner rotation-curve structure is systematically coupled to the global extent of the galaxy.

The baryonic mass and the gas fraction also show systematic couplings to the inner mass slope. For

the baryonic mass, the correlations are

$$r_{\mathrm{Spearman}} = -0.488176, \qquad p = 2.96 \times 10^{-7} \quad (Q = 1),$$

$$r_{\mathrm{Spearman}} = -0.460851, \qquad p = 6.00 \times 10^{-10} \quad (Q = 1 + 2),$$

and

$$r_{\mathrm{Spearman}} = -0.429751, \qquad p = 2.94 \times 10^{-9} \quad (Q = 1 + 2 + 3).$$

For the gas fraction, the correlation is positive:

$$r_{\mathrm{Spearman}} = 0.411626, \qquad p = 2.31 \times 10^{-5} \quad (Q = 1),$$

$$r_{\mathrm{Spearman}} = 0.357773, \qquad p = 2.75 \times 10^{-6} \quad (Q = 1 + 2),$$

and

$$r_{\mathrm{Spearman}} = 0.341599, \qquad p = 3.72 \times 10^{-6} \quad (Q = 1 + 2 + 3).$$

The galaxies were additionally divided into three size regimes. The median values of the inner mass slope are

$$\alpha_{\mathrm{median}} = 2.208303 \quad (r_{\mathrm{max}} < 16\ \mathrm{kpc},\ n = 107),$$

$$\alpha_{\mathrm{median}} = 1.940638 \quad (16 \leq r_{\mathrm{max}} < 30\ \mathrm{kpc},\ n = 32),$$

and

$$\alpha_{\mathrm{median}} = 1.448310 \quad (r_{\mathrm{max}} \geq 30\ \mathrm{kpc},\ n = 36).$$

The differences between the size regimes are statistically clear; a Kruskal-Wallis test gives

$$H = 33.137752, \qquad p = 6.37 \times 10^{-8}.$$

As a methodological control, this fully reproducible 175-galaxy reconstruction was compared with an earlier internally filtered 173-galaxy version of the same test. The earlier version used a stricter internal slope-stability construction and excluded two lower-quality $Q = 3$ objects, NGC4389 and UGC02455. In addition, the earlier slope construction was not identical to the direct reconstruction used here from $M_{\mathrm{eff}}(r) \propto g_{\mathrm{obs}}(r) r^2$.

The filtered 173-galaxy version produced stronger rank correlations. For α versus r_{max}, the values were $r_{\mathrm{Spearman}} = -0.539834$ for $Q = 1$, $r_{\mathrm{Spearman}} = -0.519145$ for $Q = 1 + 2$, and $r_{\mathrm{Spearman}} = -0.487151$ for $Q = 1 + 2 + 3$. The separation between the size regimes was also more pronounced ($H =$

44.285060, $p = 2.42 \times 10^{-10}$).

The present work nevertheless uses the 175-galaxy reconstruction as the main result, because it is generated directly from the radial SPARC observables and is therefore easier to reproduce. The fact that both variants show the same sign structure, the same monotonic ordering of size regimes, and the same qualitative coupling to system size, baryonic mass, and gas fraction supports the robustness of the result.

Within the saturation-limited field response, this result is interpreted as an expression of scale-dependent galactic organization. In this reading, smaller systems possess greater local reorganization freedom of the field. Their central structure can therefore be shaped more strongly by local saturation modulation and system-dependent inner field organization. Larger galaxies, by contrast, carry stronger large-scale field organization and approach the saturation boundary of the respective organizational regime more closely in their inner regions.

In such larger systems, additional local field tension can no longer be absorbed without limit in the center, but is more likely to be converted into redistribution, counter-coupling, or larger-scale reorganization. In the observed inner mass slope, this appears as a lower effective inner mass increase in larger systems.

The inner structure therefore does not appear as an isolated halo detail, but as a local projection of the same saturation logic that is also visible in the positive acceleration excess: baryonic organization generates a field response, while the saturation boundary determines whether this response remains centrally concentrated, scatters locally, or transitions into a broader organizational form.

The test remains methodologically cautious. It does not replace a direct density-profile reconstruction, but probes the inner rotation-curve organization through a reproducible effective mass quantity. Precisely for this reason, it is useful as an internal consistency test: the saturation-limited field response remains visible not only in the direct pointwise modeling of the positive acceleration excess, but also in the inner structure of the rotation curves.

Together, the consistency tests show that the saturation-limited field response is not confined to the direct pointwise modeling of the positive acceleration excess. It also remains visible in related representations of galactic dynamics: in the acceleration ratio, in the global baryonic mass-velocity scaling relation, and in the inner rotation-curve structure. The empirical structure is therefore broad enough to motivate the theoretical interpretation developed in the next step.

4. Theoretical Interpretation: Field Response and Saturation

The following theoretical interpretation is not intended as proof of a completed UQSH ontology. It summarizes only those minimal structural assumptions from which the empirically tested field-response form was motivated: nonlocal baryonic organization, local disk coupling, global scale dependence, and saturation-limited self-limitation. The rigor of the present work therefore lies in galaxy-separated validation, null tests, bootstrap stability, ML comparisons, and the external Bullet-Cluster stress test. The theory provides the approach; the empirical tests decide how robust this approach is.

The empirical results suggest that the positive acceleration excess in galaxies is not merely an unstructured residual of baryonic dynamics. It can be described by a structured field response parametrization. This parametrization is not purely local. It depends on the baryonic organization of the whole galaxy, the local disk structure, the radial position, and global scale quantities.

In the UQSH interpretation, this empirical structure is read as the effective response of a continuous field medium to baryonic organization (45). The central physical statement is this: the positive acceleration excess is not only a missing mass contribution in the rotation curve, but an expression of nonlocal, scale-dependent field organization with saturation.

This interpretation does not replace the standard interpretation by assertion. It provides a testable alternative structure. It must therefore satisfy exactly the features found empirically: a positive nonlocal organization contribution, a positive local disk contribution, global scale dependence, and negative saturation.

The observed acceleration is formally decomposed as

$$g_{\mathrm{obs}}(r) = g_{\mathrm{bar}}(r) + g_{\mathrm{field}}(r),$$

where

$$g_{\mathrm{field}}(r)$$

denotes the effective field response. Empirically, this quantity was determined by

$$g_{\mathrm{field,obs}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r).$$

The decisive empirical observation is that g_{field} does not grow without limit with an organization quantity. Smooth saturation improves the description of the data systematically and has a stable negative sign. This leads to the central theoretical requirement: a UQSH-like field response must not be formulated as unlimited enhancement. It must contain a finite coupling limit, organization limit, or self-limitation (45).

Schematically, such a field response can be written as

$$g_{\mathrm{field}} \sim \mathcal{R}\left[O_{\mathrm{bar}}, \Sigma_{\mathrm{local}}, r/R_d, M_{\mathrm{scale}}, R_d\right],$$

where O_{bar} denotes a nonlocal baryonic organization quantity. The empirical analysis suggests that this response is not linear and unlimited, but contains a saturation function:

$$\mathcal{R} = \mathcal{R}_0 \cdot \mathcal{A}_{\mathrm{org}} \cdot \mathcal{S}_{\mathrm{sat}}.$$

Here $\mathcal{A}_{\mathrm{org}}$ describes the enhancing effect of baryonic organization, while $\mathcal{S}_{\mathrm{sat}}$ describes a limiting effect at high effective organization strength.

In the empirical working form, this limitation appears as a negative saturation term,

$$\kappa S_{\mathrm{soft}}(X), \qquad \kappa < 0.$$

The term

$$S_{\mathrm{soft}}(X) = \log_{10}\left[1 + 10^{\log_{10} X - \log_{10} X_{\mathrm{sat}}}\right]$$

is not meant as the fundamental form of the field law. Such smooth saturation or self-limiting terms are a common phenomenological structure in effective nonlinear descriptions of fields and pattern formation, without their specific form necessarily having to be interpreted as a fundamental law (20; 21). It is a smooth phenomenological approximation of flattening. The negative sign stability of κ, however, shows that the limitation is not merely a random fit parameter. Physically, this saturation can be interpreted in several ways. First, it may describe a finite responsiveness of the field medium: baryonic organization produces a field response, but the response does not grow without limit. Second, it may describe scale coupling: the field response depends not only on the local baryonic driver, but also on how this driver is distributed across the galactic scale. Third, it may represent dynamical self-limitation: the field medium responds to structure, but at high organization density the response approaches a flattened regime.

The empirical main formula thus receives a theoretical reading that follows directly from its sign structure:

$$\log_{10} g_{\text{field}} = \text{baseline} + \text{organization} + \text{local structure} - \text{scale limitation} - \text{saturation}.$$

This interpretation is not a complete fundamental derivation. The data do not directly show the field medium itself. But they do show an empirical structure that imposes a field-mechanical reading if the positive acceleration excess is described not as a particle halo, but as a field response. The response must be nonlocal, scale-dependent, and saturation-limited.

The theoretical task is therefore not to declare the empirical formula an immediate fundamental law. It is to formulate an effective field approach whose galactic limit produces exactly the empirically required structures: nonlocal baryonic organization, local disk coupling, global scale dependence, and negative saturation.

5. A Minimal UQSH Approach as an Effective Field Equation

The empirical analysis sets three stable minimum requirements for a UQSH-like theoretical description of the positive acceleration excess. First, the response to baryonic organization must contain nonlocal structural information. Second, the local disk structure must help determine the field response. Third, the response must not grow without limit at high effective organization strength, but must contain a saturation-like limitation.

A minimal effective UQSH approach therefore does not begin with a purely local algebraic relation between g_{obs} and g_{bar}. It begins with a field state $\Phi(\mathbf{x})$ that responds to baryonic structure. Here this field state is an effective model quantity. It describes the organized response of the field medium in the galactic limit, without already fixing the complete microscopic UQSH dynamics. In this way, the approach connects to the previously formulated continuous field dynamics, where the dark matter effect is described as an effective field response to baryonic organization and saturation-limited reorganization (45). Formally, one can write

$$g_{\text{obs}}(r) = g_{\text{bar}}(r) + g_{\Phi}(r),$$

where g_{Φ} denotes the effective acceleration response of the field state Φ.

A minimal stationary working equation must therefore contain three contributions: spatial smoothing

or propagation of the field state, a baryonic source structure, and a nonlinear limitation. Schematically, this can be written as

$$\mathcal{L}_\lambda \Phi + \mathcal{N}(\Phi) = \mathcal{S}_{\mathrm{bar}}.$$

Here $\mathcal{L}_\lambda$ denotes a linear propagation or smoothing operator with characteristic coupling scale λ, $\mathcal{S}_{\mathrm{bar}}$ the baryonic source organization, and $\mathcal{N}(\Phi)$ a nonlinear saturation or self-limitation term.

A simple phenomenological realization is

$$\left(\nabla^2 - \lambda^{-2}\right) \Phi - \eta \Phi^3 = -A\, \rho_{\mathrm{bar}}^{\mathrm{eff}},$$

with

$$\eta > 0.$$

The term $\nabla^2 \Phi$ describes spatial coupling, $\lambda^{-2}\Phi$ a finite coupling length, and the nonlinear term $\eta\Phi^3$ the self-limitation of the field response. The effective baryonic source $\rho_{\mathrm{bar}}^{\mathrm{eff}}$ must not be reduced to the local baryonic density alone. It has to represent the organized baryonic structure of the system.

In a radially averaged galactic limit, the field response can then be written as a smoothed response to nonlocal baryonic organization:

$$\Phi(r) \sim \int K_\lambda(r, r')\, O_{\mathrm{bar}}(r')\, dr',$$

where K_λ is a nonlocal coupling kernel. The empirical dynamical kernels F_{dyn} can be understood as discrete, data-driven approximations of such a nonlocal operator.

The observed additional acceleration is then not identified directly with Φ, but interpreted as an effective field response,

$$g_\Phi(r) = \mathcal{G}\left[\Phi(r), \Sigma_{\mathrm{local}}(r), r/R_d, M_{\mathrm{scale}}, R_d\right].$$

In the weak or unsaturated regime, this response may appear approximately power-law-like. At high effective organization strength, however, the nonlinear term $\mathcal{N}(\Phi)$ leads to a flattening, which was empirically approximated by

$$S_{\mathrm{soft}}(X) = \log_{10}\left[1 + 10^{\log_{10} X - \log_{10} X_{\mathrm{sat}}}\right].$$

This gives the connection between empirical model and theoretical approach:

$$\text{nonlocal kernel} \quad \longleftrightarrow \quad K_\lambda(r, r'),$$

$$\text{local disk structure} \quad \longleftrightarrow \quad \Sigma_{\mathrm{local}}(r),$$

$$\text{saturation} \quad \longleftrightarrow \quad \mathcal{N}(\Phi).$$

The minimal theoretical content therefore does not lie in the concrete logarithmic regression form itself. That form is the empirical working form. The theoretical content lies in the field assumption that the positive acceleration excess arises from a nonlocal, scale-dependent, and self-limited response of the field medium.

This equation should therefore be understood as an effective limiting approach. It is not yet a unique fundamental UQSH field equation. It formulates the minimal theoretical conditions that follow from the empirical SPARC analysis: nonlocal coupling, baryonic anchoring, and saturation-limited self-limitation of the field response. Any UQSH-like field description of the positive acceleration excess must satisfy these conditions.

The next step transfers this field response perspective to gravitational lensing. The aim is not a complete lensing field equation, but a theoretical interpretation of how projected field organization can appear as effective lensing convergence.

6. Gravitational Lensing in the UQSH Framework

The transition from galactic rotation curves to gravitational lensing requires a careful interpretation of lensing convergence. In the standard picture, gravitational lensing is described as the effect of the projected gravitational mass distribution along the line of sight (12). A convergence maximum therefore marks the region of strongest projected gravitational influence. The present interpretation does not dispute this observed effect, but reads it differently: the effective lensing geometry is understood as the projection of organized field structure (44).

In this view, galaxies are not only luminous markers of a mass distribution. They are stable baryon-anchored field configurations. Each galaxy organizes the surrounding field through its baryonic structure, internal dynamics, and scale coupling. In multi-component systems, such field configurations can overlap and form an extended, coherent surface of field organization.

Gravitational lensing is therefore not treated as the effect of isolated mass points, but as the macroscopic action of a curved field structure. Incoming light-tension fronts reorganize along this field structure. The observed deflection of light is then the projected effect of this reorganization. The decisive quantity is not only the local baryonic density, but the connected field organization carried and superposed by baryonic structures.

Schematically, the observed lensing convergence can be written as the effective projection of a field-organization quantity,

$$\kappa_{\mathrm{eff}} = \mathcal{K}\left[\mathcal{F}_{\mathrm{org}}\right],$$

where $\mathcal{F}_{\mathrm{org}}$ denotes the projected baryon-anchored field organization along the line of sight, and $\mathcal{K}$ denotes the effective mapping of this field organization into an observable convergence structure. This expression is not meant as a complete lensing field equation. It is a conceptual bridge: it states that, in this interpretation, a κ-maximum marks the region of strongest effective projected field organization.

This field organization should also not be understood as an indefinitely growing quantity. When several baryon-anchored field configurations overlap, the projected field organization can grow coherently

only up to an effective organizational capacity. Beyond that point, saturation does not mean that the dynamics stop. It means a transition into redistribution, counter-coupling, or a different organizational regime. The lensing effect is therefore not simply additive, but scale-dependent and saturation-limited.

This distinguishes the present lensing interpretation from a purely local assignment of light or point mass. A κ-maximum does not have to coincide with a single optical light maximum or a single local baryonic maximum. What matters is the ordered projected field structure produced by the overlap of baryon-anchored field configurations.

This provides the theoretical bridge to the Bullet Cluster stress test that follows. The point there is not to test whether gravitational lensing is generally possible in this framework, but whether robust lensing-convergence structures in a colliding cluster geometry can be understood as projections of coherent field organization.

7. The Bullet Cluster as an External Stress Test

7.1. Methodological Role of the Bullet Cluster

After gravitational lensing has been interpreted as projected field organization, the Bullet Cluster becomes the necessary external stress test for this reading (9; 10). It concerns a different class of evidence than the SPARC rotation curves. It is not about the radial acceleration structure of individual galaxies, but about the spatial separation between projected lensing convergence, hot X-ray gas, and the galaxy-bearing environment in a cluster collision. That is precisely why it is decisive for a UQSH-like field response interpretation.

In this work, the test is not treated as a statistical main test on the same level as SPARC. It is treated as an external geometric stress test. Its task is to examine whether the basic idea of a baryonically anchored, but not gas-centered, field organization can also be formulated consistently in a colliding galaxy cluster.

The core observational finding of the Bullet Cluster is the spatial separation between the hot X-ray gas and the dominant lensing convergence structures. In the standard picture, this separation is usually interpreted as evidence for a collisionless, nonbaryonic dark matter component (9; 10). A field-organization interpretation of lensing convergence must therefore address the same separation directly. It must show that lensing convergence does not necessarily have to follow the dissipative gas maximum, but can be understood as the projection of a more coherent, galaxy-bound field organization.

Methodologically, the Bullet Cluster is not used here to replace a full dynamical simulation of the collision or a complete reconstruction of the lensing map. Instead, the test examines three limited and reproducible points: first, the positions of the dominant CANUCS κ components; second, their distances from the Chandra gas references; and third, their local optical or near-infrared environment.

The physical interpretation is deliberately separated from these measurement steps. First, the κ-gas distances and their robustness against smoothing scales are documented. Then the analysis tests whether the κ components lie in local light or near-infrared environments. Only after that is the field-organization interpretation of the finding discussed.

The Bullet Cluster test therefore has a clear role in this work. It is not a standalone proof of UQSH, but it is a necessary external stress test. A field response interpretation of the dark matter effect must pass here because the Bullet Cluster sharply poses the question of whether gravitational lensing geometry

must necessarily follow the dominant baryonic gas mass carrier. In the field-organization interpretation, the testable counterclaim is this: robust lensing convergence preferentially follows the more coherent, galaxy-bound field organization rather than the dissipatively reorganized gas maximum.

7.2. CANUCS κ Map and Chandra Gas References

The CANUCS best-fit κ map of the Bullet Cluster provides the numerical lensing-convergence layer that is compared in this work with the Chandra X-ray gas references. The underlying CANUCS/Technicolor data products include JWST and HST imaging data as well as associated catalog and lensing-model products (11). The strong-lensing model products used here are accessible through the publicly available CANUCS lensing-model page, which provides catalogs, Lenstool parameter files, and maps for the Bullet Cluster (18). Recent JWST-based studies of the Bullet Cluster further demonstrate that new NIRCam and NIRSpec data refine the lensing modeling and the spatial structure of the mass distribution (27; 16). The WCS-calibrated best-fit convergence map `bulletclu-kappa-best-50mas.fits` was used, with a pixel scale of approximately $0.05''$ per pixel.

To determine the dominant convergence structures, the κ map was evaluated over a grid of Gaussian smoothing scales between $5''$ and $15''$. This means that not only a single visual representation is considered. Instead, the analysis tests whether the dominant κ components remain stable under plausible smoothing scales.

The Chandra X-ray gas peak and the Chandra gas centroid are used as gas references (9). The test is initially purely geometrical: where are the dominant CANUCS κ components located, how stable do their positions remain under different smoothing scales, and how far are they separated from the Chandra gas references?

This quantitatively tests the core observational finding of the Bullet Cluster: the robust spatial separation between hot X-ray gas and dominant projected lensing convergence, which in the standard picture is taken as strong evidence for a collisionless, nonbaryonic dark matter component (9; 10).

Bullet Cluster: CANUCS-κ map and gas–lensing separation.
The CANUCS best-fit κ map `bulletclu-kappa-best-50mas.fits` provides a WCS-calibrated numerical convergence map with a pixel resolution of approximately $0.05''$ per pixel. After smoothing with Gaussian scales between $5''$ and $15''$, two dominant convergence components become visible.
For the eastern component, the median position is

$$\mathrm{RA} = 104.568022^\circ, \qquad \mathrm{Dec} = -55.943063^\circ,$$

with a separation from the Chandra gas peak of

$$105.37'' \pm 1.04''.$$

For the western component, the median position is

$$\mathrm{RA} = 104.654165^\circ, \qquad \mathrm{Dec} = -55.955404^\circ,$$

with a separation from the Chandra gas peak of

$$81.17'' \pm 6.42''.$$

Both convergence components therefore remain clearly separated from the Chandra X-ray gas reference across the smoothing grid used here. The eastern component is more positionally stable, whereas the western component shows a larger smoothing-dependent scatter along the extended western convergence structure.

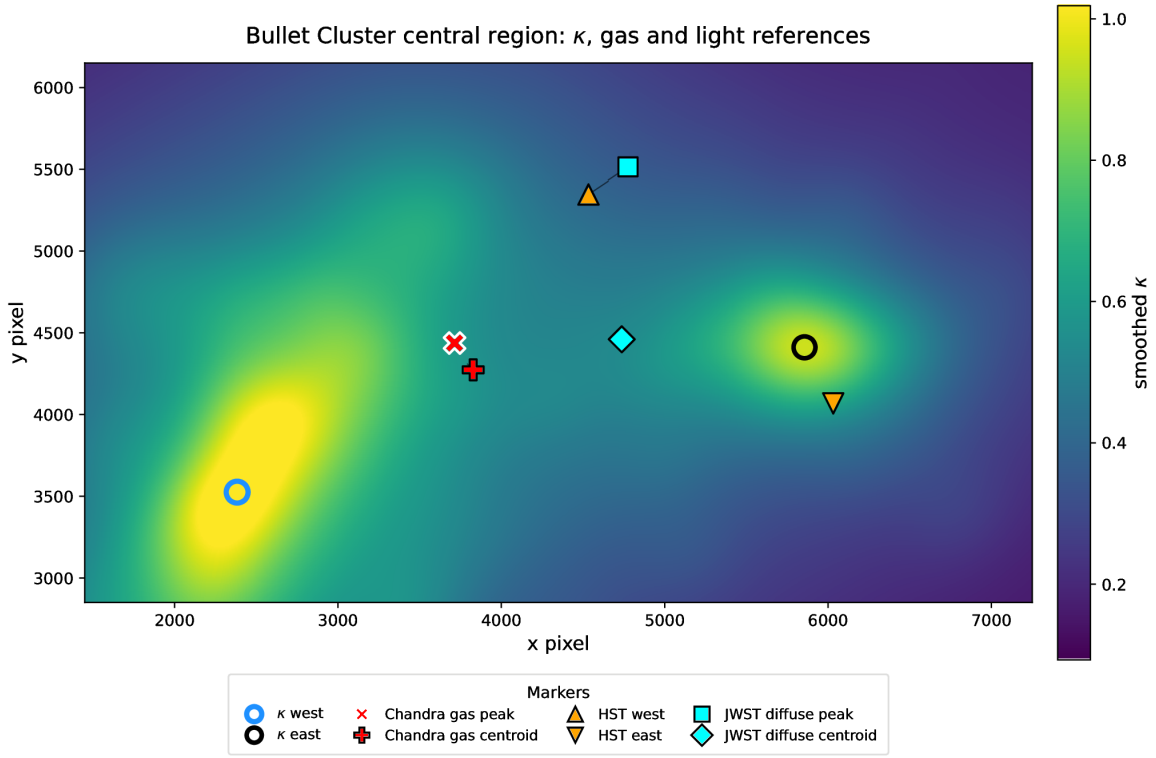


Figure 6: CANUCS best-fit κ map of the Bullet Cluster with reference markers for the two dominant convergence components, the Chandra X-ray gas peak and centroid, and the previously determined HST/JWST light proxies. The κ map is shown here after smoothing with a Gaussian scale of 10″. The figure visualizes the spatial separation between the hot X-ray gas and the dominant projected convergence structure.

The figure shows the geometric situation for a representative smoothing scale of 10″. The quantitative robustness of this κ–gas separation is tested below over the full smoothing grid.

The empirical finding is therefore not dependent on a single visual representation. The κ-gas separation remains present across the smoothing grid used.

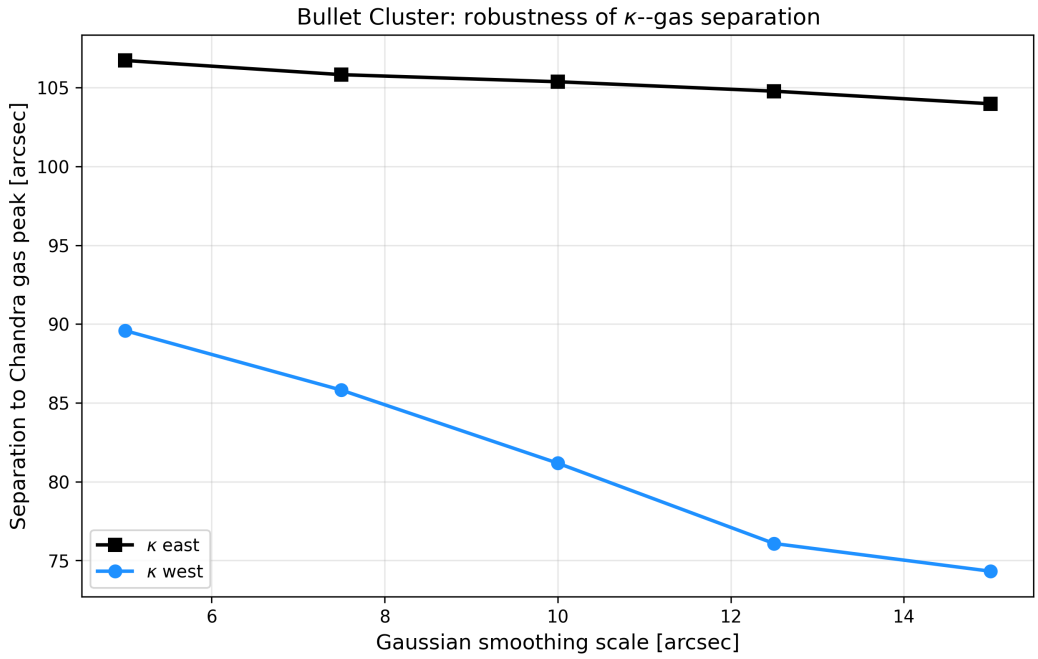


Figure 7: Robustness of the κ-gas separation against different Gaussian smoothing scales of the CANUCS best-fit κ map. The distances of the dominant western and eastern convergence components from the Chandra gas references are shown over the smoothing grid used. Both components remain clearly separated from the hot X-ray gas over the tested smoothing scales.

The smoothing-dependent robustness check is methodologically important because it separates the κ-gas separation from a single map choice. The observed separation remains present across the tested smoothing grid and therefore cannot be treated as an artifact of one visual representation or one smoothing scale.

Table 5: Robust median positions of the two dominant CANUCS best-fit κ components and their angular distances from the Chandra X-ray gas peak and centroid. The uncertainties correspond to the scatter over Gaussian smoothing scales from 5″ to 15″.

Component	RA [deg]	Dec [deg]	Gas peak [″]	Gas centroid [″]
κ west	104.654165	-55.955404	81.17 ± 6.42	80.65 ± 6.31
κ east	104.568022	-55.943063	105.37 ± 1.04	103.30 ± 1.02

This table summarizes the geometric core finding. Both dominant CANUCS κ components lie clearly outside the Chandra gas references. The eastern component is particularly positionally stable over the smoothing scales used, while the western component shows larger scatter along the extended western convergence structure. The quoted uncertainties describe the scatter of the inferred κ positions across the smoothing grid used here. They do not replace a full systematic astrometric error budget between the Chandra, HST/JWST, and lensing coordinate systems. Such systematic WCS or pointing contributions can be relevant on arcsecond scales. Since the measured κ–gas separations are of order 80″ to 105″, however, plausible astrometric systematics do not alter the qualitative core finding of a clear gas–lensing separation. The decisive point is that both components remain clearly separated

from the hot gas despite their different stability.

7.3. Local JWST/F277W Environment of the κ Components

After the geometrical κ–gas test, the local near-infrared environment of the two dominant CANUCS-κ components was examined. This step is important for the UQSH interpretation because a κ maximum need not coincide exactly with a single optical light maximum. The relevant question is instead whether the convergence structure is embedded in a coherent local galaxy or near-infrared environment, or whether it appears as an isolated, light-poor lensing peak.

For this reproducible environment check, the JWST/F277W image was used, because it covers both κ components better than the edge-limited HST/F814W region in the west. Circular apertures were evaluated for each κ component. The local background was estimated from an annulus between 450 and 650 JWST pixels and subtracted from both the aperture sums and the annular mean values.

This analysis is neither a reconstruction of stellar mass nor a substitute for a cluster member catalog. However, it is situated within the context of recent studies discussing diffuse near-infrared or intracluster light structures as useful, though not solely sufficient, tracers of cluster structure and mass distribution (29; 27). It serves only as a reproducible near-infrared environment check. In particular, no inference is made that the F277W signal is directly proportional to the total baryonic mass.

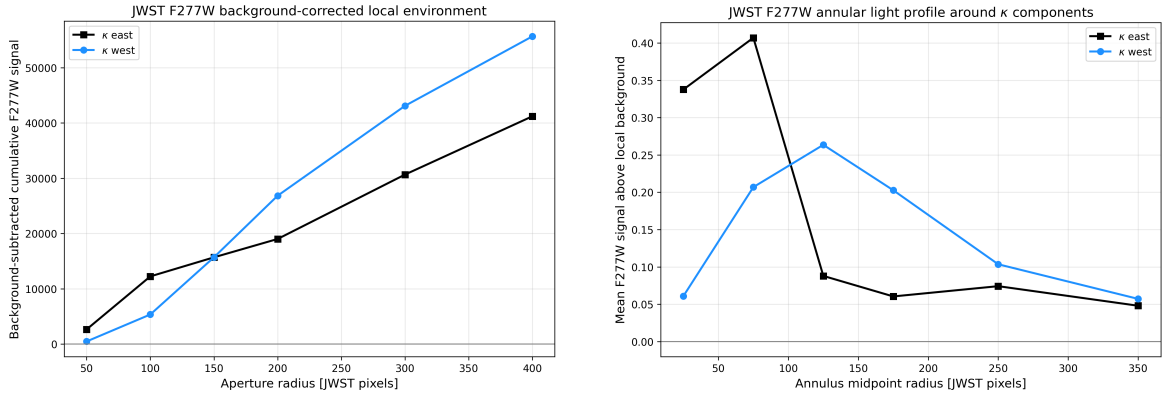
Table 6: Background-subtracted local JWST/F277W near-infrared environment of the two CANUCS-κ components. The background is estimated from a local annulus between 450 and 650 JWST pixels. The table lists the cumulative background-subtracted F277W signal, the mean aperture signal above the background, and the mean annular signal above the background.

Component	Radius [pix]	Bg.-sub. cumulative signal	Mean aperture signal	Mean annular signal
κ east	50	2654.0	0.3377	0.3377
κ east	100	12245.1	0.3897	0.4071
κ east	200	19019.0	0.1514	0.0604
κ east	400	41214.5	0.0820	0.0480
κ west	50	477.6	0.0609	0.0609
κ west	100	5354.6	0.1705	0.2069
κ west	200	26846.9	0.2137	0.2027
κ west	400	55673.4	0.1108	0.0571

The local JWST/F277W analysis shows different near-infrared environments for the two κ components. The eastern component has the stronger compact signal within small apertures. Within 50 pixels, the background-subtracted cumulative signal is approximately 2654 image units for κ east, compared with approximately 478 image units for κ west.

At larger apertures, this relation reverses. Within 400 pixels, κ west reaches approximately 55673 image units, whereas κ east reaches approximately 41215 image units. The annular profile shows the same structure: κ east is more centrally concentrated, while κ west has a broader, more distributed F277W environment over larger radii.

The previously used western HST light anchor is therefore not adopted as a robust counterpart of the western κ component. Instead, the western component is treated as a convergence structure within a more extended, measurable JWST/F277W near-infrared environment.



(a) **(b)**

Figure 8: Background-subtracted local JWST/F277W near-infrared environment of the two CANUCS-κ components. (a) Cumulative background-subtracted F277W signal as a function of aperture radius. (b) Annular F277W light profile after subtraction of the local median background. The eastern κ component dominates at small radii, whereas the western component shows a broader near-infrared profile over larger radii.

The corresponding local JWST/F277W and HST/F814W cutouts of the two κ components are not used as additional main-text figures, but are provided in the reproducible output set. There, they serve as visual checks of the local environments and help distinguish between compact light signals, broader near-infrared environments, and incomplete HST coverage in the western region.

In summary, the CANUCS-κ map, the Chandra gas reference, and the JWST/F277W environment analysis yield three separate empirical findings. First, the dominant κ components are clearly separated from the hot X-ray gas. Second, this separation remains stable across the tested smoothing scales of the κ map. Third, the JWST/F277W analysis shows that the eastern κ component is associated with a compact near-infrared signal, whereas the western component is embedded in a broader local near-infrared environment.

These findings are not interpreted below as evidence for a specific mass component. The essential point is instead that the projected convergence structure does not follow the collisional gas maximum, but remains associated with the more collisionless, structurally coherent components of the system. In the UQSH interpretation, the effective lensing response is therefore not determined by the local gas density alone, but by the spatially organized field structure carried by long-lived baryonic nodes, the light environment, and pre-existing tension organization.

7.4. UQSH Interpretation of the κ–Gas Separation

In the standard framework, the Bullet Cluster is commonly regarded as one of the strongest arguments for a collisionless dark-matter component. The observational core result is clear: the hot X-ray emitting gas is spatially separated from the dominant lensing convergence structures. Since the gas contains a large fraction of the ordinary baryonic mass of the cluster, while the lensing maxima do not coincide with the gas, the additional gravitational effect is usually attributed to a nonbaryonic, collisionless matter component.

The CANUCS analysis performed here confirms this geometrical core result. The two dominant κ components remain robustly separated from the Chandra X-ray gas references. The eastern component

is particularly positionally stable across the smoothing scales used here, while the western component shows stronger smoothing-dependent scatter along the extended western convergence structure, but remains clearly separated from the gas as well. The observation itself is therefore not disputed, but adopted quantitatively as an external stress test of the field-response interpretation.

The difference lies in the ontological reading of this result. In the UQSH interpretation, the κ structure is not necessarily read as the projection of an additional invisible particle substance, but as the projection of an effective field geometry (44). This field geometry is carried by coherent, baryonically anchored structural states. Galaxies are therefore not merely luminous tracers, but long-lived anchors of ordered field organization. During the cluster collision, their associated organizational fields can remain substantially more coherent than the hot gas component, which is collisionally decelerated, heated, and dissipatively reorganized.

The gas is not less real in this interpretation and is not outside the field ontology. It is also a baryonic field state. After the collision, however, it has a different organizational character: it is shock-heated, turbulent, and dissipative. It can still carry baryonic energy and local field excitation, but not necessarily the same long-range, coherent anchoring structure as the galaxy-bound components. The relevant distinction is therefore not between baryons and non-baryons as such, but between coherent and dissipatively reorganized baryonically anchored field states.

In this reading, the Bullet-Cluster signature is not explained by assuming that the hot gas should by itself define the dominant lensing geometry. Rather, the observed κ–gas separation is interpreted as evidence that the effective projected convergence follows the more coherent, collisionless or weakly collisional structural component of the system, rather than the dissipatively displaced X-ray gas. The galaxy-bound structures remain the relevant anchors of the ordered field geometry, while the gas traces a different, collisionally reorganized baryonic state.

The JWST/F277W environment analysis supports this interpretation in a limited but methodologically important sense. It does not replace a stellar mass reconstruction or a cluster-member catalogue. It shows, however, that the western κ component need not be treated as an empty or completely isolated lensing peak. Although the earlier manually chosen western HST light anchor is not robust because of incomplete HST/F814W coverage, the JWST/F277W data show a broader measurable near-infrared environment around the western convergence component. This is consistent with an interpretation in which not a single light peak, but the overlapping local galaxy and field environment is relevant.

This interpretation explicitly does not claim that baryonic light alone explains the full measured κ amplitude. It claims instead that visible baryonic structure, the local near-infrared environment, and pre-existing tension organization may jointly act as anchors of an effective convergence structure. In the UQSH framework, the CANUCS κ map is therefore read as the projection of a field-organized lensing response, not as a direct map of an additional invisible substance.

The Bullet Cluster thus remains a central test case. In the standard framework, it supports collisionless dark matter. In the UQSH framework, the same observation is read as evidence that lensing convergence need not follow the dissipative gas component, but can remain associated with the more coherent, galaxy-bound field organization. The open question about the microphysical nature of dark matter is therefore not avoided, but reformulated: not “which invisible substance produces the missing mass?”, but “which organizational states of the field generate the observed projected gravitational geometry?”

7.5. Distinguishable Consequences and Further Cluster Tests

The UQSH interpretation of the Bullet-Cluster signature is scientifically relevant only if it does not merely rename the same lensing structure, but formulates testable expectations for further colliding clusters. The methodological requirement is therefore clear: it must be possible to test when projected lensing convergence follows the hot gas, when it remains separated from it, and what role coherent galaxy or near-infrared environments play in this behavior.

In the standard picture, the lensing maxima essentially follow a collisionless mass component, while the hot gas lags behind because of shock processes. In the UQSH framework, by contrast, the projected lensing geometry preferentially follows those regions in which coherent, galaxy-bound field organization is preserved and overlaps. The gas is baryonic and physically real, but after the collision it is more strongly dissipative, turbulent, and thermally reorganized.

This leads to three distinguishable expectations:

1. **Robust κ–gas separation:** In strongly colliding clusters, dominant κ structures need not coincide with the X-ray gas maximum. A separation between hot gas and projected lensing convergence is therefore expected in the UQSH framework, provided that the galaxy-bound field organization remains coherently preserved.

2. **Embedding in coherent light or NIR environments:** Robust κ structures should not occur as arbitrarily lightless features. They need not coincide exactly with a single optical maximum, but they should be connected to a plausible galaxy, light, or near-infrared environment.

3. **Dissipative gas as a weaker carrier of ordered projection:** Strongly shocked, turbulent, or thermally heated gas should not necessarily determine the maximum of the ordered projected field geometry, despite containing a large baryonic mass.

The UQSH interpretation therefore shifts the focus of the test. The decisive question is not only whether lensing and gas are separated, but whether robust κ structures are simultaneously separated from the dissipative gas and embedded in coherent galaxy or near-infrared environments. The relevant observable is therefore not a single light point, but the joint geometry of κ structure, gas dissociation, and baryonically visible environment.

This expectation leads to a repeatable test protocol for further colliding galaxy clusters. For each test cluster, an astrometrically consistent data basis should first be established, consisting of a published or reconstructed lensing κ map, an X-ray map of the hot gas, and optical as well as preferably near-infrared imaging data of the galaxy-bound environment. All maps must be checked for WCS consistency.

The minimal analysis scheme comprises:

1. Determination of the dominant κ components from a high-resolution lensing map, including robustness tests with respect to smoothing scales or reconstruction variants.

2. Determination of the relevant gas references from X-ray data, in particular the gas peak, gas centroid, and, where applicable, shock-front or temperature structure.

3. Measurement of angular and projected separations between κ components, gas references, and local optical or near-infrared environments.

4. Examination of the local optical and near-infrared environment of the κ components, without prematurely equating this environment with a complete stellar mass map.

5. Comparison of the three interpretations: baryonic gas interpretation, particle dark-matter interpretation, and UQSH field-organization interpretation.

A cluster would support the UQSH interpretation if several conditions are fulfilled jointly: the dominant κ structures remain stable across plausible smoothing scales or reconstruction variants, are robustly separated from the dissipative gas, and are at the same time connected to coherent galaxy, light, or near-infrared environments. Exact coincidence with a single light maximum is not required.

A critical finding for the UQSH would instead be a stable, massive κ peak that persists in several independent lensing reconstructions, is clearly separated from the gas, and at the same time has no plausible optical or near-infrared baryonic environment. Equally critical would be a systematic coincidence of the most robust κ maxima with the dissipative gas maximum in strongly colliding systems, despite the galaxy distribution being clearly separated from it.

In this way, the Bullet-Cluster interpretation is not formulated as a retrospective image interpretation, but as a falsifiable cluster program. Further systems must show whether the same combination recurs: robust κ–gas separation, embedding of the convergence structures in coherent galaxy or near-infrared environments, and no systematic binding of the strongest lensing maxima to the dissipative gas maximum. Particularly relevant next test cases are systems with complex collision geometry and robust multi-wavelength data, such as El Gordo (30; 39) or MACS J0717 (32; 33).

8. Falsifiable Predictions and Next Tests

The preceding sections show that the saturation-limited field response can be tested not only within SPARC rotation curves, but also provides a distinguishable reading for lensing systems such as the Bullet Cluster. The decisive point is therefore to turn the field-organization interpretation into concrete predictions rather than leaving it as a retrospective interpretation.

The following tests are formulated so that they can support, weaken, or falsify the field response interpretation. The issue is not a single qualitative agreement, but repeatable patterns in independent data sets: rotation curves, MDAR/RAR residual structure, BTFR consistency, core-cusp-related inner structure, and colliding galaxy clusters.

Table 7: Summary of the most important falsifiable test domains of the saturation-limited field-response interpretation. They are not meant as confirmation of a completed theory, but as conditions under which the field-response interpretation must either hold up empirically or fail.

Test domain	UQSH expectation	Critical finding
SPARC rotation curves	The positive acceleration excess remains predictable, galaxy by galaxy, from baryonic structure, scale parameters, and saturation.	The model performance disappears in independent galaxy samples or is reproduced by shuffle/null tests.
MDAR/RAR residual structure	Residuals contain regime- and structure-dependent information and are not purely stochastic.	RAR/MDAR residuals show no stable coupling to radius, surface density, system size, or saturation indicators.
BTFR and global scaling	The global mass–velocity structure remains consistent with the local field response.	The local field response improves point-wise data but systematically destroys global BTFR consistency.
Inner structure	Core-cusp-related inner slopes couple to global system size and field organization.	Inner slopes are completely independent of global extent, structural regime, and field-response parameters.
Colliding clusters	Robust κ structures follow coherent galaxy/NIR environments rather than dissipative gas.	Stable κ maxima repeatedly occur without baryonic or NIR structural environments and can only be described as an independent massive additional component.

In detail, five priority test lines follow.

1. Independent rotation-curve samples. The most important direct test is the application of the same saturation-limited field response formula to independent rotation-curve data sets outside the SPARC sample used here. The model parameters must not be freely refitted for each new galaxy. They must be tested in a galaxy-wise validation scheme. A strong finding in favour of UQSH would be a stable prediction of the positive acceleration excess with performance comparable to that in SPARC. It would be critical if the predictive power disappeared outside SPARC or could be retained only through renewed free parametrization.

2. Structure of MDAR/RAR residuals. A second test line concerns the residual structure in MDAR and RAR space. In the field-organization interpretation, deviations from the empirical RAR should not be completely random. They should correlate systematically with structural quantities such as radius, local baryonic surface density, system size, saturation indicators, or inner field organization. The field response interpretation used here would be supported if such residuals reproduced the same regime dependencies in independent samples.

A critical finding would be a case in which the RAR/MDAR residuals show no stable coupling to galactic structure parameters after measurement uncertainties are taken into account. In that case, the

saturation-limited field response might still work as an empirical parametrization, but its interpretation as an expression of a real field-organized additional structure would be clearly weakened.

3. BTFR and global mass-velocity scaling. A third test line concerns the compatibility of the local field response with global scaling relations. The saturation-limited field response must not only improve individual radial acceleration points. It must remain compatible with the baryonic Tully-Fisher relation. In the field-organization interpretation, this is expected because local field organization and global mass-velocity scaling are not independent. They are different projections of the same galaxy-wide organization structure.

A supportive finding would be that independent galaxy samples show that the field response improves local acceleration data without systematically increasing the BTFR scatter or distorting the global mass-velocity relation. A critical finding would be that local model performance is achieved only at the cost of losing global BTFR consistency. In that case, the field response would be more like a local fitting formula than a consistent description of galactic organization.

4. Inner structure and core-cusp-related slopes. A fourth test line concerns the inner rotation-curve structure. In the field-organization interpretation, the inner effective mass slope should not appear exclusively as a local property of a central halo or feedback process. It should at least partly couple to global system properties. The relation found in this work between inner slope and maximum radial extent of the rotation curve provides a first empirical indication of this.

A supportive finding would be that independent samples confirm that core-cusp-related inner structures are systematically connected to global galaxy extent, baryonic organization, or field response parameters. A critical finding would be that the inner slopes remain fully independent of global structural quantities after controlling for measurement resolution and sample selection. That would clearly weaken the field-organization interpretation of scale-dependent inner field organization.

5. Colliding clusters and lensing geometry. A fifth test line concerns colliding galaxy clusters. In the UQSH interpretation, projected lensing convergence in such systems should not primarily follow the dissipative X-ray gas, but the more coherent, galaxy-bound structure carriers and their local optical or near-infrared environment. The decisive point is not the exact coincidence of a κ maximum with a single light maximum, but the robust relation between convergence structure, galaxy environment, NIR structure, and gas dissociation.

A supportive finding would be that further systems such as El Gordo or MACS J0717 show robust κ structures that repeatedly remain separated from dissipative gas and are at the same time embedded in coherent galaxy or near-infrared environments. A critical finding would be a stable, massive κ peak that remains present in independent lensing reconstructions, is clearly separated from the gas, and at the same time has no plausible optical or near-infrared baryonic environment. Such a finding would significantly weaken the field-organization interpretation of lensing convergence as a field-organized, baryonically anchored structure.

Taken together, these five test lines define an empirical test program for the saturation-limited field response interpretation. UQSH is not presented as a complete theory, but as a falsifiable framework. It must show that the same saturation structure visible in rotation curves also remains consistent with global scaling relations, inner galaxy structure, and lensing geometries of colliding clusters.

The decisive point is repeatability. Individual good fits, individual illustrative lensing overlays, or

individual cluster examples are not enough. The interpretation becomes robust only if independent data sets repeatedly show the same coupling between baryonic structure, saturation, field response, and projected convergence geometry.

The approach therefore has clear failure conditions. It would be weakened if saturation, nonlocal structural coupling, and cluster κ environment are not reproducible in independent data.

9. Extended UQSH Interpretation

The previous sections have presented the empirically tested structure: the positive acceleration excess has a nonlocal, scale-dependent, and saturation-limited form; this structure is validated with galaxy-separated testing, controlled by null tests, and compatible with an external Bullet-Cluster stress test. Only on this basis is the broader UQSH interpretation now discussed.

The following sections should therefore not be understood as additional empirical proofs, but as a theoretical interpretation of the structure tested above. They describe which properties a UQSH-like field medium would have to possess if the empirically found field response is indeed an expression of a deeper field dynamics.

9.1. Minimal Properties of the UQSH Field Medium

In the Universal Quantum Foam Hypothesis, a distinction is made between a prior Qu-ground Q and the physically readable field reality Φ. Here, Q is not a measured quantity, not a field value, and not an object within space and time. What is physically accessible is only Φ: the effective field reality within which motion, tension, curvature, coupling, saturation, and relaxation become measurable or indirectly inferable. Formally, this distinction can be written compactly as

$$Q \rightarrow \Phi \rightarrow B(\Phi),$$

where $B(\Phi)$ denotes the motion and reorganization of the physically effective field reality. The present work makes no direct measurement claim about Q. All empirical statements refer to observable effects within Φ. In this context, the term Qu-foam does not denote a material-like medium inside space, but the scale-relative organizational and transitional structure through which the prior Qu-ground Q appears as the physically effective field reality Φ.

The empirical analysis of the rotation curves, the saturation-limited field response, the core-cusp-related inner structure test, and the Bullet Cluster stress test indicate which properties a UQSH-like effective field medium must at least possess if the dark-matter phenomenology is to be interpreted as a field response. These properties are not meant as a complete ontology of the Qu-foam, but as physical minimum requirements for the field reality denoted in this work by Φ. The term field projection does not refer here to a virtual or merely mathematical environment, but to the physically effective, measurable or indirectly inferable manifestation of the underlying Qu-ground.

Several minimal properties of Φ follow from the dark-matter phenomenology. They can be grouped into three classes: coupling and anchoring, saturation and reorganization, and projection, causality, and scale structure.

Coupling and anchoring.

1. **Nonlocal coupling.** The field response must not depend only on the local density value. It must include the organized baryonic structure of the system as a whole. The positive acceleration excess is therefore not understood as a purely local residual term, but as the expression of a system-wide field organization that acts beyond the immediate location.

2. **Baryonic anchoring.** Visible matter generates or stabilizes field organization, but is not identical with the entire effective field geometry. Galaxies, disk structure, baryonic mass distribution, and local organizational states act as anchors around which the field response forms. The dark-matter phenomenology therefore remains baryonically bound without being reducible to visible matter alone.

3. **Distinction between field organization and observable coupling.** Not every field-effective structure must appear directly as light, matter, or radiation in our observational regime. A field organization can be physically effective, carry tension, and contribute to the effective gravitational geometry without becoming immediately electromagnetically visible. This separates existence, field effect, and visibility coupling.

Saturation and reorganization.

4. **Finite local organizational capacity.** Additional field tension cannot be concentrated locally without limit. The field response must therefore possess an effective saturation limit. In the empirical analysis, this property appears through the negative saturation term: organization amplifies the effect, but only up to a limited local regime.

5. **Relaxation and redistribution.** Saturation does not mean the end of the dynamics. When the local organizational density reaches a limiting regime, the field tension does not disappear, but is redistributed, counter-coupled, relaxed, or transferred into a different organizational regime. This property explains why enhanced field action need not grow arbitrarily in the center, but can pass into flatter or more broadly distributed organizational forms.

6. **Coherent superposition and network-forming capacity.** Freely propagating tension fronts must not be considered only in isolation. They must be able to continuously cross, overlap, and thereby generate a large-scale, dynamic tension network. The decisive point is not the persistence of a single front as a permanent object, but the ongoing reorganization of many overlapping contributions from which an effective field topology emerges.

Projection, causality, and scale structure.

7. **Dynamical-geometrical projection capacity.** The same field organization must be able to appear both dynamically, as a positive acceleration excess in rotation curves, and geometrically, as projected lensing convergence. Rotation curves, MDAR/RAR structure, BTFR, inner rotation-curve structure, and cluster lensing are therefore not treated as separate phenomena, but as different projections of the same field-organizing logic.

8. **Finite limiting rate of reorganization.** The field must possess a finite limiting rate of state change. This limiting rate is the speed of light c. In the UQSH framework, c therefore denotes not only the propagation speed of electromagnetic radiation, but the maximum causal transfer rate of state changes within Φ, valid scale-invariantly. Light is the freely propagating mode of motion of this field projection,

$$L = B_{\text{free}}(\Phi),$$

whereas matter, inertia, and proper time appear only where field motion is bound, stabilized, and readable within a regime. This creates a direct conceptual connection to general relativity: there, too, c is not merely a property of light, but the universal limiting speed of causal influence. The UQSH adopts this limiting structure, but interprets it as the maximum reorganization rate of the physically readable field reality Φ.

9. **Scale-relative or fractal-like organization.** The field must be able to carry similar organizational roles across different scales. This does not mean that an exact mathematical fractal dimension of the Qu-foam has already been determined. Rather, it means that coupling, tension, binding, saturation, relaxation, and reorganization can occur not only microscopically, but also galactically and possibly cosmologically. Fractality is therefore not an additional measured quantity in the present preprint, but a structuring theoretical expectation.

9.2. UQSH Interpretation of Radiation

For this interpretation, it is important that radiation in the UQSH framework is not understood as a small object moving through a medium. It is instead read as a freely propagating mode of motion of the physically effective field projection Φ. In this sense, light is not a special case of a different ontology, but the most visible form of this radiation for us: a locally generated, spherically propagating tension or boundary front of the field.

In the standard picture, radiation is usually described as an electromagnetic wave or, in the quantum picture, as a photon. The UQSH interpretation does not call into question the measurability of photons, spectral lines, energy quanta, frequencies, intensities, or temporal structures, but it does question their usual ontological reading. Here, the photon is not understood as the primary building block of radiation, travelling as a small object through space in finished form, but as a local coupling or exchange event of an extended field motion.

From this perspective, the standard description remains an extremely successful observational and computational description. However, it need not already capture the deeper carrier structure of radiation completely. This possible gap is decisive for the UQSH: if radiation is understood as a modulated field front, redshift, intensity decay, time dilation, field tension, and later gravitational organization can be read as connected projections of the same field motion. The detector then does not measure the entire extended front, but only the small part that couples locally and temporally to its material structure.

A local field tension propagates, in the idealized case, isotropically at the limiting rate c. The front carries not only a propagation motion, but also an internal order. In the observational regime, this internal order appears as frequency, spectrum, line structure, intensity, and temporal signature. Radiation therefore has two distinguishable aspects: the propagating tension front itself and the modulation of this front. The modulation is not a second object alongside the front, but its internal, readable structure.

In previous UQSH work, spectral shift, time dilation, and intensity decay were formulated accordingly as three observable projections of a common path-dependent scaling of such boundary dynamics (46). The present work does not adopt this interpretation as a completed cosmology, but as an illustrative field interpretation of how radiation can carry frequency, energy, spectrum, and intensity.

The following idealized illustration first shows the spherical propagation of individual radiation-field fronts.



Figure 9: Idealized spherical propagation of radiation-field fronts as expanding tension structures in the UQSH. Such tension fronts arise from local field reorganizations, for example when bound field states release energy, relax, or couple to another organizational regime. In detection, the entire tension front is not captured; only a very small locally and temporally coupled part of its extended propagation is measured. The spacing of successive fronts illustrates frequency. Wave-like behavior arises from the ordered sequence of arriving tension fronts, or from the internal modulation of the propagating boundary dynamics. The Light Bubble is therefore idealized not as a filled sphere, but as a propagating tension-and-relaxation skin.

An individual front shell can be imagined intuitively as an expanding envelope. The same field motion is distributed over an ever larger surface as propagation continues. As a result, the local tension and coupling density of the front decreases. The front does not disappear, but its locally measurable effect per coupling region becomes weaker.

This de-densification affects several observable properties at once. The intensity decreases because a detector couples only to a smaller effective part of the extended front than at the source origin. The locally readable energy and coupling content decreases because the same field motion is distributed over a larger effective region. At the same time, the internal modulation of the front is read with a larger pattern width: the distances between successive fronts or modulation phases appear larger, and the observed frequency decreases. In this UQSH reading, redshift is therefore not a destruction of the radiation order, but a scale-relative stretching of its internal modulation.

The decreasing local tension density affects, in this reading, not only electromagnetic readability, but also the field-curving gravitational effect of the front. The decisive factor is not the total front as an extended structure, but the local tension difference that a coupling region actually experiences. A dense front carries a stronger local field deformation; a strongly de-densified front, by contrast, produces only a weak local change of the field state. Its curvature effect therefore does not disappear, but is de-densified below electromagnetic detectability into a weaker or differently coupled subregime.

The spectrum is an internal modulation of the radiation front. It is not the front itself, but the ordered structure with which the front couples to bound field states and is read out by them. In this UQSH interpretation, spectral lines do not arise as material markings on a small particle, but as resonant modulation responses of bound field organizations. After coupling, this modulation order continues to propagate with the front.

Illustratively, this can be compared to a transparent soap bubble. The bubble itself is hardly visible; it becomes visible only when its surface shimmers in color. These colors are not the bubble itself, but the local reading of its surface modulation. Similarly, the observer does not register the entire radiation front, but the state of its modulation at the moment of resonance coupling: with a certain intensity, a certain energy content, and a certain redshift.

This reading must not be equated with the classical tired-light hypothesis. In historical tired-light models, redshift was interpreted as an energy loss of photons during their propagation through space, without requiring metric expansion (40; 41). Such models are under strong observational pressure, in particular from supernova time dilation and Tolman surface-brightness tests (42; 43).

The UQSH interpretation, by contrast, does not claim that light simply "becomes weaker" or loses its energy as an isolated particle. The radiation front remains present as field motion, but with spherical propagation it distributes its tension, coupling, and modulation density over a larger surface. Redshift, intensity decay, and decreasing local coupling capability are therefore not simple loss processes, but different readings of a scale-relative redistribution of the same propagating tension front.

This point is essential for the UQSH interpretation of the dark-matter effect. In this reading, radiation is not only a carrier of signals, but also a carrier of field tension and modulation. Regions with high radiation density, high baryonic activity, or strongly bound field organization may therefore enter a saturation-near regime more rapidly. The energy or curvature content of dense radiation fields then contributes to local field organization until the effective coupling or organizational capacity is reached. With decreasing baryonic anchoring, falling radiation intensity, or progressive de-densification, the coherent field response also becomes weaker. In this reading, the dark-matter effect therefore depends not only on visible mass, but also on the ability of baryonic structures to organize radiation, tension, and field modulation coherently.

The role of density is central because a denser tension front can carry more internal modulation. These modulations may themselves be understood, within the subregime, as smaller tension or coupling fronts, whose exact degree of densification and dynamics were theoretically addressed in earlier UQSH work on the field ontology of light, Light Bubbles, and the Gravity-Light interpretation (47; 48; 49). High-energy radiation, in particular gamma radiation, corresponds in this illustrative UQSH reading to an especially densely organized and strongly modulatable front. It could therefore lie close to a maximally effective saturation regime without fully suppressing the dynamics.

Conversely, radiation that has travelled very far and has become strongly de-densified appears only in a weak coupling regime. In this interpretation, the cosmic microwave background would be a limiting example of a very far-travelled, strongly scaled radiation form that is still readable by our

detectors. Further de-densification increasingly carries it out of our electromagnetic coupling range. This interpretation is meant as a UQSH-internal illustration of why radiation, saturation, baryonic anchoring, and dark-matter phenomenology must be considered together.

Formally, this idea can be written minimally by means of a path-dependent scaling factor $\sigma(\gamma)$. For a propagation path γ, one may write schematically

$$\nu_{\mathrm{obs}} = \frac{\nu_{\mathrm{src}}}{\sigma(\gamma)}, \qquad \Delta t_{\mathrm{obs}} = \sigma(\gamma)\,\Delta t_{\mathrm{src}}, \qquad I_{\mathrm{obs}} = I_{\mathrm{src}}\,K(\gamma).$$

Here, $\sigma(\gamma) > 1$ denotes a de-densification or stretching of the internal boundary dynamics. The observed frequency decreases, temporal signatures are stretched, and the intensity is reduced through a path-dependent coupling factor $K(\gamma)$. Spectral shift, time dilation, and intensity decay therefore appear not as three independent processes, but as three observable readings of the same scale-relative reorganization of the radiation front (46).

Thus, the radiation interpretation in this work is tied to its role in the UQSH field picture: radiation is the propagating front; frequency, spectrum, energy content, intensity, and redshift are different modulations and coupling readings of that front. The following point is therefore not the complete cosmology of radiation, but its possible role in cumulative field tension, saturation, and baryonically anchored field organization.

On macroscopic scales, such fronts do not occur in isolation. Many sources continuously generate spherical tension fronts, which cross, overlap, and form a dynamic superposition network.

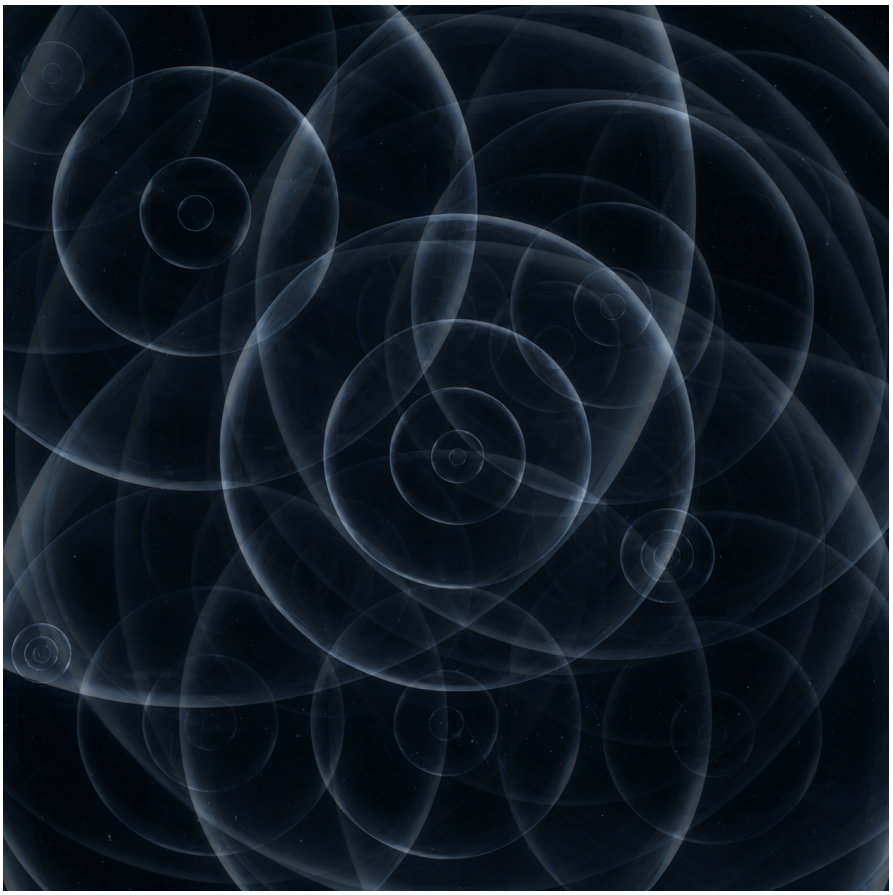


Figure 10: Schematic representation of overlapping spherical tension fronts. The figure shows an idealized limiting situation in which tension fronts propagate isotropically at the limiting speed c and form a dynamic overlap network. In the UQSH interpretation, the continuous superposition of such fronts in the quantum regime contributes to a cumulative field tension, or to a field-curving and therefore gravitationally effective response. In real systems, the sources of the fronts are themselves in motion; overlaps therefore do not arise simultaneously or at fixed locations, but in spatially and temporally shifted coupling regions. The representation is therefore not to be understood as a literal reconstruction of individual fronts, but as a conceptual visualization of the superposition principle.

In real astrophysical systems, this geometry is substantially more complex. The fronts arise on the relevant microphysical coupling scale. The sources of the tension fronts are in motion, so overlaps do not arise simultaneously or at fixed locations, but in spatially and temporally shifted coupling regions. Moreover, because of the enormous number of overlapping contributions, an individual front is generally not isolable. What is observable is therefore not the tension fronts themselves, but their interactions with matter, for example through emission, absorption, scattering, or reflection. The figure deliberately reduces this situation to the underlying principle of overlapping propagation fronts.

9.3. UQSH Interpretation of the Field

It is important that Φ must not be confused with an ordinary visible medium. The field is not like water, air, rubber, or a stretched net. Such media display only individual, regime-bound expressions of more general field properties: water shows continuity, wave formation, and relaxation; air shows pressure propagation and permeability; elastic bodies show restoring behavior and nonlinear load

limits; stretched nets show tension coupling and nonlocal load distribution. In the UQSH framework, however, these visible properties are not models of the field. They are later, material, and regime-bound manifestations of properties that are understood more deeply as properties of the field reality itself.

For this reason, asking why the field is not wet like water, fibrous like a net, or material like air is a category error. Wetness, visibility, fiber structure, and mechanical friction are macroscopic properties of bound matter within Φ. They are not properties of the field reality as such. Φ carries states, tensions, reorganizations, couplings, saturations, and relaxations; it does not itself appear as a material substance inside space.

This also gives the distinction from the historical ether. The classical luminiferous ether was conceived as a substance-like carrier medium inside space, relative to which an observer could move. Interferometer experiments such as the Michelson–Morley experiment therefore searched for a direction-dependent light travel time, that is, for an ether wind. In the UQSH framework, the null result of such experiments is expected: if Φ is not a mechanical substance and not a moving carrier medium with its own rest frame, then there is no ether wind against which the Earth could move. The experiment therefore rules out a classical substance-like light carrier, but not the possibility of a stationary field reality whose local states carry light, gravitation, and field tension.

UQSH is also not identical with Wheeler's quantum foam. Wheeler's quantum foam mainly denotes the idea of microscopic spacetime fluctuations at extremely small scales. The UQSH Qu foam, by contrast, is not merely a Planck-scale picture and not a collection of small material bubbles. The term denotes a scale-relative organization of field reality: on microscopic scales, local reorganizations may appear foam-like or cell-like; on macroscopic scales, the same principle appears as the propagation of field tension, relaxation fronts, gravitational waves, galactic field organization, or lensing geometry.

The distinction can therefore be summarized as

classical ether = moving carrier medium for light inside space,

Wheeler's quantum foam = microscopically fluctuating spacetime structure,

UQSH Qu foam = scale-relative reorganization of a stationary field reality.

These properties remain partly speculative because the field medium is not observed directly. They are not arbitrary, however: they are constrained by the phenomena tested in this work. A UQSH-like field medium must carry nonlocal baryonic organization, be baryonically anchored, allow local saturation, relax and redistribute tension, project its field organization both dynamically and geometrically, possess a finite causal reorganization rate, form scale-relative organizational roles, distinguish between field action and visibility coupling, and allow a dynamic overlap network of free tension fronts. This minimum structure connects the SPARC results, the core-cusp-related inner structure test, the saturation structure, the light-front interpretation, and the Bullet Cluster stress test.

9.4. UQSH Interpretation of the Dark-Matter Effect

The empirical analysis in this work is motivated by a speculative UQSH interpretation of the dark-matter effect. This interpretation is not assumed as an already established foundation, but as a theoretical

framework from which testable expectations for the observable dynamics can be derived. The rigor of the present work therefore does not lie in claiming the underlying medium as proven, but in the galaxy-separated validation of the field-response structure motivated by it.

Against the background of the field-medium properties described above, the dark-matter effect does not appear in the UQSH as an additional invisible matter density, but as the macroscopic effect of cumulative field curvature. Many spherically propagating tension fronts overlap, carry local tension and deformation of the medium, and can be spatially organized by baryonic structures. Individual free tension fronts do not yet generate a stable macroscopic dark-matter effect. Only their persistent superposition, baryonic anchoring, and nonlocal organization can appear as a coherent additional gravitational effect.

Baryonic structures have a double role in this interpretation. On the one hand, they are sources of field excitation, because stars, gas, compact objects, disk structures, and active baryonic processes continuously generate radiation and tension fronts. On the other hand, they act as anchoring structures, because they spatially order and stabilize existing field tension. The dark-matter effect therefore does not arise from field excitation alone, but from baryonically organized field excitation.

This distinction is particularly relevant for baryon-poor, extremely diffuse galaxies, or those discussed as being dark-matter-deficient (22; 23; 24; 26). In such systems, field excitation may still be present, without the baryonic structure being sufficiently coherent to form a stable macroscopic field organization. A weak or apparently absent dark-matter effect would therefore not necessarily indicate the absence of field excitation, but rather insufficient baryonic anchoring and organization of the existing field tension.

In addition, the field medium is subject to saturation. Cumulative field curvature cannot grow without limit in a local region. Once the local absorption capacity of the medium is reached, additional field tension must transition into flattening, counter-coupling, or nonlocal redistribution. The UQSH interpretation of the dark-matter effect therefore contains three central elements: tension-front excitation, baryonic source and anchor organization, and saturation-limited nonlocal reorganization.

In this interpretative reading, the empirical field term of the present analysis can be written as the galactic limiting case of an organized field response,

$$g_{\mathrm{field}} \equiv g_{\mathrm{DM}}^{\mathrm{UQSH}} = -\nabla\Phi_{\mathrm{DM,eff}}, \qquad \Phi_{\mathrm{DM,eff}} = \Phi_{\mathrm{rad}} + \Phi_{\mathrm{nl}}.$$

Here Φ_{rad} describes the cumulative field excitation from overlapping, baryonically generated tension fronts. The term Φ_{nl} describes the nonlocal redistribution of this field tension once local saturation sets in. The observed dynamics then receives the UQSH reading

$$g_{\mathrm{obs}} = g_{\mathrm{bar}} - \nabla(\Phi_{\mathrm{rad}} + \Phi_{\mathrm{nl}}).$$

This equation is not used in the present work as a fully derived fundamental law. It serves as a compact theoretical interpretation of the empirically validated field response. The data analysis shows that the positive acceleration excess has a reproducible nonlocal, baryonically coupled, and saturation-limited structure. The UQSH interprets this structure as the macroscopic effect of cumulative field curvature: tension fronts move the medium, baryonic structures organize this motion, and saturation limits and redistributes the resulting field response.

10. Discussion

The present results do not point to a single isolated fitting result, but to a recurring structure across several observational levels. In the SPARC rotation curves, the positive acceleration excess can be predicted galaxy by galaxy by a saturation-limited field response; at the same time, this structure remains compatible with MDAR/RAR, BTFR, inner rotation-curve organization, and the external Bullet-Cluster stress test.

The central point of discussion is therefore not whether the observed dark-matter phenomena exist. These phenomena are fully accepted. What is at issue is their ontological interpretation: are they the expression of an additional nonbaryonic matter component, or can they be read as effective projections of a structured, saturation-limited field response?

10.1. Empirical Significance of the Field-Response Structure

The results presented in this work should not be understood as a free empirical curve fit to galactic rotation curves. The fitted parametrization is the tested galactic limiting form of a more general field-response structure: baryonic organization excites a nonlocal field response, this response is locally limited by saturation, and the resulting acceleration excess becomes observable in rotation-supported galaxies as the dark-matter effect.

This interpretation is supported by several features of the analysis. The response is not determined by baryonic acceleration alone, but also by radial structure, local baryonic surface density, galactic scale parameters, and a saturation term. The sign structure of the coefficients remains stable under galaxy-separated bootstrap resampling, galaxy-shuffle tests destroy the model performance, and flexible machine-learning baselines do not provide a fundamentally different explanation under the same validation logic and comparable constraints on the functional form. The decisive point is therefore the organized relation between baryons, scale, radius, and field response.

The empirical parametrization is thus not the starting point of the interpretation, but the observed galactic limiting form of the proposed field-response mechanism.

10.2. Relation to RAR, MOND, and ΛCDM

The present analysis does not replace the empirical RAR, but uses it as a central reference structure. The RAR describes the observed coupling between baryonic and dynamical acceleration with high accuracy (2; 3). The saturation-limited field-response model starts from a different point: it does not directly model the full observed acceleration, but the positive excess $g_{\mathrm{obs}} - g_{\mathrm{bar}}$. The RAR is therefore not disputed, but used as a limiting and comparison case within a more general field-response description.

Compared with MOND, the difference is that no fundamental modification of the acceleration law is postulated here (4). The empirical proximity to the RAR instead arises from a structured, baryonically anchored additional response with saturation. The relevant point is therefore not only a universal acceleration scale, but the question of whether the positive acceleration excess can be predictively reconstructed from galaxy structure, scale information, and nonlinear field response.

Compared with ΛCDM, the difference does not lie in the observational phenomena themselves. Flat rotation curves, RAR/MDAR (2; 3), BTFR structure (5), core-cusp-related tensions (6; 7), and

lensing offsets in cluster collisions (9; 10) are not disputed in this work. The difference lies in the interpretation of the additional gravitational effect. In the standard framework, it is explained by a nonbaryonic, weakly collisional matter component. In the UQSH framework, it is tested whether the same effect can be described as the effective projection of a saturation-limited, baryonically anchored field organization.

In addition to these established reference frameworks, there are current works that describe rotation curves with empirical correction terms, FDM/SFDM halo profiles, or multi-regime halo models. The present work differs from them by validating the positive acceleration excess as a target variable in its own right, galaxy by galaxy, by controlling it with galaxy-shuffle and bootstrap tests, and by linking the saturation-limited field response additionally to MDAR, BTFR, inner rotation-curve structure, and Bullet-Cluster lensing.

The present work therefore does not stand in isolation as a purely empirical parametrization. It builds on a previously formulated first-principles-based UQSH approach in which the dark-matter effect was developed as a consequence of continuous field dynamics, propagating tension fronts, baryonic organization, local saturation, and nonlocal redistribution. The contribution of the present analysis is to test this theoretically motivated structure empirically on real rotation curves, inner structure indicators, and an external lensing stress test.

10.3. Limitations of the Present Analysis

The present work does not provide a complete microphysical field theory of the UQSH. The saturation-limited field response used here is an effective empirical parametrization that tests whether the positive acceleration excess can be described as structured, predictive galaxy by galaxy, and compatible with further observational levels. Good empirical model performance therefore does not yet constitute proof of the underlying field ontology.

A second limitation concerns the data basis. The SPARC analysis relies on rotation curves, baryonic decompositions, and fixed mass-to-light assumptions. Systematic uncertainties in distance, inclination, gas contribution, stellar populations, or bulge-disk decomposition can affect the resulting accelerations. The quality classes used here reduce this problem, but do not remove it completely.

A third limitation concerns the Bullet Cluster. The CANUCS κ map, the Chandra gas references, and the JWST/F277W environment analysis allow a geometrical stress test of the UQSH interpretation, but they do not replace a complete cluster mass modeling. In particular, no complete stellar mass map, no dynamical merger model, and no independent lensing reconstruction from the raw data are presented here. The Bullet-Cluster part should therefore be understood as a methodological and interpretive stress test, not as a completed cluster theory.

A direct information-criterion comparison with separately fitted halo profiles remains reserved for future work and is not required for the aim of the present analysis. The goal of this work is to test whether the positive acceleration excess admits a reproducible saturation-limited field-response interpretation.

Finally, it remains open whether the empirical saturation structure can be derived from a more fundamental UQSH field equation. The present work shows a recurring, falsifiable structure. Deriving this structure from a microphysically precise field dynamics remains a necessary next step.

10.4. Next Steps

The limitations just described lead to four immediate next steps. First, the saturation-limited field response must be applied to independent rotation-curve data. The decisive point is that the same structure should not only work within SPARC, but should also persist in external samples under galaxy-separated validation. Particularly relevant are high-resolution rotation-curve and mass-model data for dwarf galaxies, such as those provided by LITTLE THINGS (8), as well as current HI-based rotation-curve and mass-model samples from MIGHTEE-HI (17).

Second, additional baryonic proxies should be tested systematically. WISE- and GALEX-based quantities (34; 35) are particularly relevant because they can provide complementary information on stellar populations, star formation, and the radiation environment. A viable field-response model must show whether such proxies reduce the residual structure in a controlled way without destroying the saturation structure itself.

Third, the cluster test must be extended to further systems. After the Bullet Cluster, El Gordo and MACS J0717 are particularly suitable next test cases. The same scheme of κ-robustness, X-ray gas references, and optical or near-infrared environment should be applied to these systems.

Fourth, stronger theoretical formalization is required. The empirical saturation must be derived from a field equation or at least from a more tightly motivated effective action principle. Only then can it be decided whether the observed structure is merely a robust empirical parametrization or the limiting case of a deeper UQSH field dynamics.

11. Conclusion

The present work examines whether dark-matter phenomenology can be described as a reproducible and empirically testable field interpretation. The central question is whether the positive acceleration excess $g_{\mathrm{obs}} - g_{\mathrm{bar}}$ can be understood as the expression of a nonlocal, scale-dependent, and saturation-limited field response to baryonic organization. The starting point was not the assumption of a completed UQSH field theory, but the test of whether such a structure can be modeled predictively, galaxy by galaxy, and consistently with further observational levels.

The central empirical result is that the saturation-limited field-response model reconstructs the positive acceleration excess with high accuracy. In the high-quality $Q = 1$ sample, it reaches practically the level of the empirical RAR; in the extended $Q = 1 + 2$ and $Q = 1 + 2 + 3$ samples, it lies slightly below the RAR RMSE. The decisive point is not only this numerical proximity to the RAR, but the stable structure of the description: nonlocal organization acts as an amplifying contribution, local disk structure contributes positively, global scales limit the response, and the saturation term remains negative.

This structure is supported by several control steps. The galaxy-shuffle null test shows that the real assignment between galaxy structure and field excess is not reproduced by randomly reassigning target values between galaxies. The galaxy-separated bootstrap confirms the stability of the central sign structure, and generic machine-learning baselines do not outperform the interpretable saturation model under the same validation logic. In addition, the Bullet-Cluster stress test shows that the dominant CANUCS κ structure is robustly separated from the Chandra X-ray gas, and that at least the western κ component need not be treated as an empty lensing peak, but lies within a broader JWST/F277W near-infrared environment.

Taken together, this yields a coherent empirical structure: baryonic organization amplifies the field response, local disk structure provides an additional contribution, global scales limit the response, and a saturation-like flattening prevents unlimited local growth. Saturation does not mean the end of the dynamics, but the transition from further local amplification to redistribution, counter-coupling, or a changed organizational regime. This structure forms the empirical core of the UQSH interpretation of a continuous, nonlocal, and self-limited field response.

The interpretation of gravitational lensing extends this reading to projected field organization. In the UQSH framework, a lensing κ maximum does not necessarily mark a single local mass or light maximum, but the region of maximally effective projected field organization. In this sense, the Bullet Cluster serves as an external stress test: the observed κ–gas separation is not denied, but read as an indication that the dominant lensing convergence can remain connected to more coherent galaxy-bound structures rather than following the dissipatively reorganized gas maximum.

At the same time, the scope of the interpretation remains clearly limited. The analysis replaces neither a full dynamic cluster simulation nor external tests using independent rotation curve data. It does show, however, that the UQSH-related idea of a nonlocal and saturation-limited field response can be formulated not only qualitatively, but as a concrete, reproducible, and falsifiable empirical structure.

The next step is therefore the external testing of these predictions: independent rotation-curve data, additional WISE- and GALEX-based structure and stellar-population proxies, renewed machine-learning controls, and further colliding clusters with lensing, X-ray, and near-infrared data.

Code and Data Availability

The analysis code, validation notebooks, processed output tables, figures, and reproducibility material for this work are available in the public GitHub repository

`https://github.com/ukshinrexhepi-cloud/UQSH_DarkMatter_Project`

The repository contains the SPARC validation workflow, the saturation-limited field-response analysis, the galaxy-shuffle and bootstrap controls, the machine-learning baseline comparisons, and the Bullet-Cluster lensing validation material. The repository is intended as a reproducibility package for the empirical results reported in this paper. External observational data products remain subject to their original data releases and citations. The reproducibility material is available in the project repository (50).

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and all conclusions of this work remain with the author.

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